UME

Organisation / Business Operating System

Business Cases, Use Cases & User Workflows

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UME Product & Design · Internal

16 Business Cases · 48 Use Cases · 32 User Workflows

Solo Entrepreneur → SMB → Enterprise → Franchise DAO

Document Overview

This document defines the business cases, use cases, and user workflows for the UME Organisation Operating System (OrgOS) platform. It covers every major platform module and persona, from a solo entrepreneur bootstrapping their first business to an enterprise operator managing a franchise DAO of 50 organisations.

Structure

Part	Content	Sections
Part I — Business Cases	Why each type of organisation and persona uses UME. ROI rationale, problem statements, and value delivered for 16 distinct business scenarios across 6 org archetypes.	BC-01 through BC-16
Part II — Use Cases	Functional use cases: actor, goal, preconditions, main flow, alternate flows, exceptions, and post-conditions. 48 use cases covering all major OrgApp capabilities.	UC-01 through UC-48
Part III — User Workflows	End-to-end workflow specifications: trigger, actors, steps, decision points, system automations, cross-module events, and success criteria. 32 workflows covering complete business processes.	WF-01 through WF-32

Org Archetypes Covered

Archetype	Example	Modules Active	Org Scale
Solo Entrepreneur	Freelance consultant, independent contractor	Finance, Legal, CRM, Work	1 person
Small Service Business	Dry cleaning shop, hair salon, accountancy practice	Finance, HR, Legal, CRM, Ops, Marketing	2–20 people
Growth SMB	Multi-location service chain, growing tech startup	Full module suite + AI Agents	20–200 people
Enterprise / Multi-Entity	Group of companies, holding structure, corporate HQ	Enterprise tier + Multi-entity + DAO	200+ people
Franchise Network	Franchise brand + independent franchisee operators	DAO + DLT + OrgDNA broadcast	N independent orgs
Investment Fund / DAO	Investment vehicle + portfolio companies	DAO + DLT + Investment module	N portfolio orgs

Part I: Business Cases

Business cases establish the organisational rationale for using the UME platform. Each business case describes a specific type of organisation, the problems they face without UME, the value UME delivers, and the measurable outcomes expected. Business cases are the foundation for all use cases and workflows that follow.

16 Business Cases across 6 Org Archetypes

BC ID	Title	Archetype	Primary Driver
BC-01	Solo Entrepreneur: First Business Formation	Solo	Compliance & legal setup
BC-02	Freelancer: Integrated Client & Finance Management	Solo	Invoice-to-cash & CRM
BC-03	Small Dry Cleaning Business: Full Operational OS	SMB	Operations + HR + CRM + Finance
BC-04	SMB: Replacing a Fragmented Tool Stack	SMB	Consolidation & cost reduction
BC-05	Professional Services Firm: Client Delivery & Compliance	SMB	CRM + legal + HR
BC-06	Retail Business: Multi-Channel Sales & Inventory	SMB	Sales + inventory + operations
BC-07	Growth Company: AI-Augmented Organisation	Growth SMB	AI OrgExecs + automation
BC-08	Multi-Location Business: Single OS Across Sites	Growth SMB	Multi-entity + consolidation
BC-09	Corporate Group: Consolidated Multi-Entity Management	Enterprise	Multi-entity finance + GRC
BC-10	Enterprise: OrgExec Governance & AI Agent Deployment	Enterprise	ODD + AI agent governance
BC-11	Nonprofit: Grant, Compliance & Stakeholder Management	Enterprise	Restricted funds + GRC
BC-12	Franchise Network: DAO Formation & Brand Governance	Franchise	DAO + OrgDNA + smart contracts
BC-13	Franchise Operator: Running Under a DAO	Franchise	OrgOS + DAO membership
BC-14	Investment Fund: Portfolio Company OS Management	Investment DAO	DAO + DLT + investment module
BC-15	Supply Chain DAO: Buyer-Supplier Network Coordination	DAO	Inter-org contracts + DLT
BC-16	Enterprise Spinout: OrgDNA Cloning & Launch	Enterprise	OrgDNA clone + rapid launch

BC-01 — Solo Entrepreneur: First Business Formation

BC-01 First Business Formation

Primary Persona	New Entrepreneur / Founder
Org Type	Solo — any industry
Core Problem	A first-time entrepreneur has a business idea but no framework for making it legally real, financially visible, or operationally manageable. They face compliance requirements they do not understand, tools they cannot afford, and processes they have to invent from scratch.
Value Delivered	UME provides a single guided wizard that walks the entrepreneur from idea to operating business: entity formation, chart of accounts, compliance calendar, initial contracts, and basic CRM — all in one session, with no prior business knowledge required.

Problem Statements

1. Entity registration and jurisdiction compliance requirements are opaque and intimidating for first-timers.
2. Accounting setup (chart of accounts, period management, tax categories) requires specialist knowledge.
3. Missing a filing deadline in the first year can result in penalties that destroy early-stage cash flow.
4. No affordable, integrated way to track both finances and clients in one place.

Value Delivered

< 60 min	Time from account creation to fully configured, legally-aware OrgOS
0 missed filings	Compliance calendar auto-generated from entity type + jurisdiction; reminders 14 days before each deadline
$0 accountant fees	COA auto-generated from business type; standard bookkeeping requires no specialist in Year 1

Success Criteria

Entity details entered; jurisdiction detected; compliance calendar populated; COA created; first invoice sent — all within one session.

Modules Activated

Module	Used For
Legal & Compliance (Chombo)	Entity registration, jurisdiction detection, compliance calendar generation
Finance & Accounting	Chart of accounts, invoicing, basic bookkeeping
CRM / Sales	Customer list, first quote/invoice creation

Dashboard	Daily overview of cash, open invoices, and upcoming filings

BC-02 — Freelancer: Integrated Client & Finance Management

BC-02 Freelancer Operations

Primary Persona	Freelancer / Independent Contractor
Org Type	Solo — professional services
Core Problem	A freelancer manages clients across email, spreadsheets, and separate invoicing tools. Time tracking is manual, invoices are created ad-hoc, expense tracking is after-the-fact, and tax prep requires reconstructing a year of scattered records.
Value Delivered	UME integrates CRM (client and project pipeline), finance (invoicing, expenses, VAT/tax tracking), and work management (task and time tracking) into one OS. Tax year-end is a report run, not a reconstruction project.

Key Metrics

4.2 hrs/wk	Average admin time saved vs. fragmented tool stack (industry benchmark)

100%	Invoice-to-payment tracked in one system — no chasing records across tools

Modules Activated

Module	Used For
CRM / Sales	Client pipeline, project tracking, quotes, invoices
Finance & Accounting	Expense recording, payment tracking, VAT/tax categories, reports
Work Management	Task lists, time tracking, deliverable tracking per client
Legal & Compliance	Contractor agreements, NDA storage, annual filing reminders

BC-03 — Small Dry Cleaning Business: Full Operational OS

BC-03 Dry Cleaning Business — Full OS

Primary Persona	Business Owner (Bob & Alice)
Org Type	SMB — Dry Cleaning / Local Service
Core Problem	Bob and Alice run a dry cleaning business using a mix of paper order tickets, a basic POS, a spreadsheet for payroll, and their accountant's software for taxes. Nothing talks to anything else. They have no real-time view of their business health, miss compliance deadlines, and spend 6+ hours per week on admin.
Value Delivered	UME provides a single OrgOS that runs every aspect of the business: order management (CRM kanban), payroll-linked HR, automated compliance calendar, finance with real-time P&L, equipment monitoring as OrgCPUs, and an AI scheduling agent as an OrgExec — all connected and consistent.

Before vs. After

Domain	Before UME	After UME
Order Management	Paper tickets, verbal status updates, no customer notification	Digital order kanban: Received→Cleaning→Ready→Collected; auto SMS on status change
Finance	Monthly spreadsheet sent to accountant; no real-time view	Live P&L, auto-categorised transactions, one-click payroll journal entries
HR & Payroll	Paper timesheets, manual payroll calculation	Digital timesheets, automated payroll calculation, payslips generated automatically
Compliance	Accountant reminds them; sometimes miss deadlines	Chombo filing calendar: all state/federal obligations auto-populated; 14-day alerts
Equipment	Notice problems when machines break	OrgCPU IoT status: machine health, maintenance schedules, alerts before failure
Scheduling	Bob decides daily by gut; staff call to check shifts	AI Scheduling OrgExec: optimises assignments based on order queue and staff availability

Quantified Value

6 hrs/wk	Admin time saved — freed for serving customers
$2,400/yr	Estimated accountant fee reduction from self-managed books
0 penalty filings	Compliance calendar prevents missed state and federal deadlines

BC-04 — SMB: Replacing a Fragmented Tool Stack

BC-04 Tool Stack Consolidation

Primary Persona	Business Owner / Operations Manager
Org Type	SMB — any industry
Core Problem	A typical SMB uses 8–12 SaaS tools: accounting (Xero/QuickBooks), HR (BambooHR), CRM (HubSpot/Salesforce), project management (Monday/Asana), payroll (Gusto), e-sign (DocuSign), storage (Google Drive), and more. Each tool has its own login, pricing, and data model. Data is duplicated, integrations break, and staff waste hours switching contexts.
Value Delivered	UME replaces the entire fragmented stack with one OrgOS where all data lives in one system and modules share a consistent data model. The event bus automatically propagates changes (a new hire in HR triggers IT provisioning, payroll setup, and legal document creation).

Tool Stack Replacement Map

Tool Being Replaced	UME OrgApp Equivalent	Integration Preserved
Xero / QuickBooks	Finance & Accounting OrgApp	Import historical COA; Stripe payment sync
BambooHR / HiBob	HR & People OrgApp	Import employee records; payroll integration
HubSpot CRM	Sales & CRM OrgApp	Import contact + deal history
Monday / Asana	Work Management OrgApp	Import open tasks and projects
Gusto / ADP Payroll	HR Payroll sub-module	Payroll run stays inside OrgOS
DocuSign	Legal & Compliance vault + e-sign integration	Existing signed docs archived in vault
Google Drive (partial)	Document vault across all OrgApps	Drive remains for unstructured files; org docs in vault
Slack (partial)	Organisation Communications OrgApp	Slack integration for notifications remains

$1,800–$4,200/yr	Estimated SaaS cost savings from tool consolidation (8-tool stack → 1)

BC-05 — Professional Services Firm: Client Delivery & Compliance

BC-05 Professional Services Firm

Primary Persona	Managing Partner / Practice Manager
Org Type	SMB — Law, Accounting, Consulting, Architecture
Core Problem	Professional services firms must simultaneously manage client engagements (delivery, billing, contracts), regulatory compliance (professional licences, data protection, client confidentiality), staff resourcing and utilisation, and financial performance — typically across 3–5 disconnected systems.
Value Delivered	UME provides a unified OS where client engagements, time recording, invoicing, HR utilisation, compliance obligations, and partner financials all live in one place and drive each other automatically.

Core Value Drivers

Real-time utilisation	HR module + Work module provide live staff utilisation: billable vs. non-billable hours by person, team, and project
Engagement profitability	Finance + CRM integration: P&L per client engagement calculated automatically from time records and invoice data
Compliance confidence	Chombo tracks professional licence renewals, CPD requirements, PI insurance, and data protection obligations

BC-06 — Retail Business: Multi-Channel Sales & Inventory

BC-06 Retail Business Operations

Primary Persona	Retail Owner / Operations Manager
Org Type	SMB — Retail, E-commerce
Core Problem	A retail business managing both a physical store and online channel faces perpetual inventory reconciliation problems, disconnected sales reporting, manual fulfilment tracking, and supplier order management done via email and spreadsheets.
Value Delivered	UME's Operations/ERP and Sales modules provide a unified inventory pool shared across all channels, automated reorder points triggering purchase orders, and real-time sales dashboards showing margin by SKU and channel.

Key Capabilities

Capability	Description
Unified Inventory	Single inventory pool visible across all sales channels; adjustments from any channel update the shared pool
Automated Reorder	Low-stock alerts generate purchase order drafts automatically; vendor lead times calculated into reorder point
OrgCPU Integration	POS terminal registered as OrgCPU; sales events flow directly into inventory and finance modules
Supplier Management	Vendor records with preferred terms; PO generation; goods receipt matching; accounts payable tracking

BC-07 — Growth Company: AI-Augmented Organisation

BC-07 AI-Augmented OrgOS

Primary Persona	CEO / COO / Operations Lead
Org Type	Growth SMB — tech, services, logistics
Core Problem	A fast-growing company cannot hire fast enough to keep pace with operational demand. Scheduling, customer communications, demand forecasting, and compliance monitoring all require human attention that the team does not have bandwidth for.
Value Delivered	UME's OrgExec model allows AI agents to be deployed as first-class OrgExecs with formally defined ODD contracts. AI agents handle scheduling, customer comms, and compliance monitoring autonomously — escalating to humans only when their ODD constraints require it.

AI OrgExec Deployment Examples

AI OrgExec	ODD Capabilities	ODD Constraints	Value
Scheduling Agent	Read order queue, staff availability; Write schedule assignments; Notify staff of shifts	Max 200 executions/hr; no changes affecting >3 staff without human approval	Eliminates manual scheduling; optimises order throughput
Customer Comms Agent	Read order status; Send SMS/email notifications; Update CRM contact notes	Pre-approved message templates only; no pricing changes; escalate complaints	Reduces customer service workload by 60-80% for status updates
Compliance Monitor	Read all OrgApp data; Write risk alerts; Escalate overdue filings	Read-only on all modules; cannot modify records; alert-only	24/7 compliance monitoring without dedicated compliance staff
Demand Forecaster	Read sales history, inventory; Write reorder suggestions	Suggestions only — no autonomous purchasing; human approval required	Reduces stockouts and overstock; improves cash conversion cycle

BC-08 — Multi-Location Business: Single OS Across Sites

BC-08 Multi-Location Operations

Primary Persona	Group Owner / Regional Manager
Org Type	Growth SMB — retail, food & bev, services
Core Problem	A business with 3–10 locations operates each site as a separate entity with separate tools, giving management no consolidated view of group performance. Cross-site comparisons require manual data extraction and reconciliation.
Value Delivered	UME's multi-entity model allows each location to be a legal entity within a single OrgOS, sharing the same module configuration. Consolidated financials, headcount, and KPIs are available instantly across all entities.

Multi-Entity Value

Instant consolidation	Group P&L and balance sheet computed in real-time across all entities — no month-end spreadsheet reconciliation
Standardised operations	All locations run the same OrgApps with the same processes — OrgDNA ensures consistency
Location benchmarking	Side-by-side KPI comparison across locations drives performance conversations with data

BC-09 — Corporate Group: Consolidated Multi-Entity Management

BC-09 Corporate Group Management

Primary Persona	CFO / Group Finance Director / Board
Org Type	Enterprise — holding structure, group of companies
Core Problem	A corporate group managing 5–20 legal entities faces consolidated financial reporting that takes weeks, separate compliance obligations per entity, intercompany transaction complexity, and no single system of record for the group.
Value Delivered	UME provides each entity with its own OrgOS instance within a group DAO, with real-time consolidated financial reporting, intercompany reconciliation, and a group-level risk and compliance dashboard.

Enterprise Value

Function	Without UME	With UME
Consolidated financials	4–6 weeks of manual spreadsheet work per month-end	Real-time consolidated P&L and balance sheet; intercompany eliminations automated
Compliance management	Each entity's compliance managed independently; gaps inevitable	Group compliance calendar: all entities' obligations in one view; central oversight
Intercompany transactions	Manual journals; reconciliation errors; audit risk	Intercompany transactions recorded in both entity OrgOS instances simultaneously via DAO event bus
Group risk view	No unified risk picture; separate risk registers if any	Group risk dashboard aggregates KRIs from all entity OrgOS instances

BC-10 — Enterprise: OrgExec Governance & AI Agent Deployment

BC-10 Enterprise OrgExec Governance

Primary Persona	CIO / CISO / COO
Org Type	Enterprise — any regulated industry
Core Problem	A large enterprise deploying AI across operations has no standardised framework for governing AI agent behaviour. Each AI deployment has its own ad-hoc controls. Audit trails for AI actions are incomplete. The board cannot attest to AI governance.
Value Delivered	UME's ODD (OrgDeviceDriver) protocol provides a standardised, auditable governance framework for every AI agent deployed as an OrgExec. Every AI action is logged, every constraint is formally defined, every escalation is recorded — the same framework used for human employees.

Governance Value

100% AI actions audited	Every OrgExec action — human or AI — produces an immutable audit record with the same structure and retention
Board-attestable AI governance	ODD contracts are formal, versioned documents that can be reviewed and approved by audit committee
Zero AI privilege escalation	ODD capability model: AI agents can only do what their ODD explicitly permits; no implicit access

BC-11 — Nonprofit: Grant, Compliance & Stakeholder Management

BC-11 Nonprofit Operations

Primary Persona	Executive Director / Finance Manager
Org Type	Nonprofit, NGO, charity
Core Problem	A nonprofit must manage restricted funds (each grant has specific spending rules), demonstrate compliance to funders, report to trustees, manage volunteers, and handle regulatory obligations — all with limited staff and minimal technology budget.
Value Delivered	UME's Finance module supports restricted fund accounting with separate tracking per grant. The GRC module manages compliance obligations. The HR module tracks volunteers as OrgExecs with appropriate ODDs. The Portal module surfaces reporting dashboards for trustees.

Nonprofit-Specific Capabilities

Capability	Description
Restricted Fund Accounting	Each grant is a separate fund; expenditure tagged to fund; automatic alerts if fund is approaching limit or nearing end date
Trustee / Board Portal	Board members access a read-only dashboard showing financials, compliance status, and programme delivery metrics
Volunteer OrgExec Management	Volunteers registered as Human OrgExecs with VOLUNTEER-ODD; activity tracked; DBS/clearance document vault
Funder Reporting	Reporting module generates grant expenditure reports in funder-required format from tagged transaction data

BC-12 — Franchise Network: DAO Formation & Brand Governance

BC-12 Franchise Network DAO

Primary Persona	Franchise Brand Owner / Franchisor
Org Type	Franchise — any consumer-facing industry
Core Problem	Bob and Alice want to franchise their dry cleaning business. They need: a way to share their operational playbook with franchisees, automatic royalty collection, brand standards enforcement, and group governance — without micromanaging each franchisee.
Value Delivered	UME's DAO module allows Bob and Alice to form a franchise DAO. Their OrgDNA becomes the brand package. Smart contracts automatically collect royalties. Brand standards compliance is checked automatically. Franchisees operate their own fully independent OrgOS instances while remaining aligned to the parent brand.

DAO Franchise Value

Capability	Description
OrgDNA as Franchise Kit	Export the parent OrgOS config as OrgDNA; franchisees clone it to spin up an operational dry cleaning OrgOS in <30 minutes
Automatic Royalty Collection	Smart contract: 6% of monthly revenue auto-debited from franchisee treasury to parent treasury on period close event — no invoicing, no chasing
Brand Standards Enforcement	Smart contract monitors franchisee OrgDNA version; if >1 major version behind parent, triggers cure period and governance escalation
Collective Governance	Major brand decisions (price changes, new service lines, marketing spend) put to governance vote; quorum of franchisees required
Group Performance Dashboard	Parent org sees consolidated revenue, orders, and NPS across all franchisees in real-time

BC-13 — Franchise Operator: Running Under a DAO

BC-13 Franchise Operator Experience

Primary Persona	Franchisee Business Owner
Org Type	Franchise — operating under a parent DAO
Core Problem	A franchisee wants the benefits of an established brand and proven operating model without being locked into a rigid central system that ignores local market realities. They want operational autonomy within brand guidelines.
Value Delivered	Each franchisee gets a full, independent OrgOS — every module, every capability. The DAO connection simply adds: automatic royalty payment, OrgDNA brand updates, and governance voting. The franchisee is operationally sovereign while being brand-aligned.

Franchisee Experience

Full OrgOS ownership	Franchisee controls all their data, operations, and staff — DAO only governs brand compliance and royalties
< 30 min onboarding	Clone parent OrgDNA → adjust entity details and jurisdiction → launch OrgOS
Zero royalty admin	Smart contract handles royalty calculation and transfer automatically — franchisee just runs the business

BC-14 — Investment Fund: Portfolio Company OS Management

BC-14 Investment Fund & Portfolio DAO

Primary Persona	Fund Manager / General Partner
Org Type	Investment Fund — VC, PE, family office
Core Problem	An investment fund manages 8–15 portfolio companies, each with different tools, reporting formats, and data quality. Getting portfolio-level financial and operational data requires weeks of manual data collection and normalisation.
Value Delivered	Each portfolio company runs its own OrgOS. The fund forms a DAO with all portfolio companies. Portfolio-level financial consolidation, KPI dashboards, and risk monitoring are real-time — no manual data collection.

Portfolio DAO Value

Real-time portfolio view	All portfolio company financials, headcount, and KPIs visible to fund managers in real-time via DAO dashboard
Standardised reporting	All portfolio companies using same OrgOS framework produce comparable data automatically — no format normalisation
OrgDNA as investment thesis	Fund publishes its operational best-practice OrgDNA to portfolio companies; accelerates operational improvement post-investment

BC-15 — Supply Chain DAO: Buyer-Supplier Network

BC-15 Supply Chain DAO

Primary Persona	Supply Chain Director / Procurement Manager
Org Type	Enterprise — manufacturing, retail, distribution
Core Problem	A buyer organisation manages 20–50 suppliers, each with separate systems, ordering processes, and data formats. Purchase orders are emailed. Lead times are guessed. Quality issues are discovered late.
Value Delivered	Buyer and key suppliers form a supply chain DAO. Smart contracts govern SLAs, automatic PO generation, and payment terms. Shared inventory visibility means the buyer can plan against supplier stock levels in real-time.

Supply Chain DAO Value

Function	Value
Automatic PO generation	Low-stock event in buyer's inventory triggers PO smart contract; supplier receives confirmed order instantly
SLA monitoring	Smart contract tracks delivery performance; SLA breach triggers automatic penalty clause per contract terms
Shared demand signal	Buyer's demand forecast shared with key suppliers via DAO event bus; suppliers plan production accordingly
Payment automation	Goods receipt confirmed in buyer's OrgOS triggers payment smart contract; no manual accounts payable processing

BC-16 — Enterprise Spinout: OrgDNA Cloning & Rapid Launch

BC-16 OrgDNA Clone & Rapid Launch

Primary Persona	Corporate Development / Integration Manager
Org Type	Enterprise — M&A, spinout, new venture
Core Problem	A corporate group wants to spin out a new business unit as an independent entity, or rapidly launch a new market entry. The traditional approach requires months of tool procurement, IT setup, and process design.
Value Delivered	UME's OrgDNA clone capability allows the parent group's proven operational configuration to be exported, adapted, and used to provision a fully operational new OrgOS in under 30 minutes.

Spinout Process

Step	Action	Time
1	Export parent OrgDNA (excluding transactional data)	5 min
2	Adjust: new entity name, jurisdiction, COA accounts, compliance packs	15 min
3	Provision new OrgOS from adapted OrgDNA	< 2 min automated
4	Invite founding team as OrgExecs with appropriate ODDs	5 min
5	New OrgOS is live and fully operational	< 30 min total

< 30 minutes	Time from decision to operational OrgOS — vs. 3–6 months for traditional IT setup

Part II: Use Cases

Use cases define the functional interactions between actors and the UME OrgOS platform. Each use case specifies the actor, goal, preconditions, numbered main flow, alternate/exception flows, and post-conditions. Use cases are grouped by module domain.

Group	UC IDs	Module Domain
A — Authentication & OrgOS Setup	UC-01 – UC-04	Auth, Provisioning, OrgDNA
B — Finance & Accounting	UC-05 – UC-10	Finance
C — HR & People	UC-11 – UC-16	HR, OrgExec, ODD
D — Legal & Compliance	UC-17 – UC-21	Legal / Chombo
E — Sales & CRM	UC-22 – UC-26	CRM, Sales
F — Operations & Supply Chain	UC-27 – UC-31	Operations / ERP
G — Risk & GRC	UC-32 – UC-35	Risk / GRC
H — OrgSystem (ODD, SDO, OrgApps)	UC-36 – UC-41	OrgExec, ODD, SDO, OrgDNA
I — DAO & DLT	UC-42 – UC-46	DAO, DLT, Smart Contracts
J — Marketing & Work	UC-47 – UC-48	Marketing, Work Management

Group A — Authentication & OrgOS Setup

UC-01 Register a New Account & Launch OrgOS [Auth / OrgOS Provisioner]

Actor	New User (any persona)
Goal	Create a UME account, configure an OrgOS from a template, select modules, set entity details, and launch a working Organisation Operating System.
Pre-conditions	User has no existing UME account. User has a valid email address.
Post-conditions	User has a fully operational OrgOS with selected modules running, compliance calendar populated, COA generated, and initial roles assigned. OrgDNA v1.0.0 committed.

Main Flow

1	User navigates to ume.io and clicks "Create my OrgOS".
2	User enters full name, organisation name, work email, and password. Accepts Terms of Service.
3	System sends a 6-digit OTP to the email address.
4	User enters OTP. System verifies and activates account.
5	System presents OrgOS Template Gallery. User selects an industry template (e.g., Dry Cleaning).
6	System presents Module Configurator pre-populated for the chosen template. User toggles desired modules; dependency graph updates in real-time.
7	User enters legal entity type, jurisdiction, and tax ID. System fetches applicable compliance obligations.
8	User optionally invites team members, assigning roles and OrgExec types.
9	User clicks "Launch OrgOS". System provisions OrgKernel, all selected OrgApps, COA, compliance calendar, and initial ODDs.
10	System displays OrgOS boot sequence with real-time step-by-step progress.
11	Boot completes. User lands on Executive Dashboard.

Alternate / Exception Flows

Invalid OTP	OTP entry fails: system shows attempt count. After 3 failures, resend is offered. After 10, account locked and support email sent.
Existing email	System detects existing account for email. Prompts user to sign in instead or reset password.
Module dependency conflict	If user disables a required module, system highlights dependents and prevents disabling until dependencies resolved.
Jurisdiction not supported	System flags jurisdiction as "limited compliance pack" and shows available coverage, allowing user to proceed with warning.

UC-02 Sign In with Multi-Factor Authentication [Auth]

Actor	Any existing user
Goal	Authenticate securely and resume work from last active screen.
Pre-conditions	User has an active UME account with MFA enabled.
Post-conditions	User has an authenticated session. Activity log entry created. Last-active screen displayed.

Main Flow

1	User enters email and password on sign-in screen.
2	System validates credentials. If invalid, increments failure counter (locks after 10).
3	System prompts for MFA: TOTP code or push notification.
4	User enters TOTP code or approves push notification.
5	System validates MFA. If valid, creates session token.
6	System restores user to their last active screen and org context.

Alternate / Exception Flows

Forgot password	User clicks "Forgot password". System sends reset link to registered email. Reset link valid for 15 minutes.
MFA device lost	User clicks "Use backup code". System accepts a single-use backup code, invalidates it, and prompts MFA re-enrolment.
Multiple orgs	If user is a member of more than one OrgOS, an org selector is shown before dashboard load.
SSO login	User clicks Google/Microsoft/SAML SSO button. Identity provider handles authentication. System creates session on successful callback.

UC-03 Export and Clone an OrgDNA Package [OrgDNA Manager]

Actor	Organisation Owner / Admin
Goal	Export the current OrgOS configuration as an OrgDNA package and use it to provision a new OrgOS instance.
Pre-conditions	User has Owner or Admin role. OrgOS is in a stable, validated state.
Post-conditions	New OrgOS instance is live. OrgDNA v1.0.0 committed to new instance. Parent-child OrgDNA lineage recorded.

Main Flow

1	User opens OrgSystem → OrgDNA Manager.
2	User clicks "Export OrgDNA". System serialises kernel config, all OrgApp settings, ODD templates, COA, compliance packs, RBAC policies, and KPI definitions.
3	System runs validation: checks all dependencies satisfied, no circular refs, no incomplete configs.
4	System generates versioned OrgDNA package (TOML + JSON). User downloads or stores to OrgDNA vault.
5	User clicks "Clone to New OrgOS". Wizard opens: user enters new org name, adjusts jurisdiction and entity details.
6	User optionally modifies COA codes or compliance pack for the new jurisdiction.
7	User confirms. System provisions new OrgOS instance from cloned OrgDNA in background.
8	Boot completes for new instance. User receives email with access details for new OrgOS.

Alternate / Exception Flows

Validation fails	OrgDNA validation finds an incomplete OrgApp config. System lists specific issues with links to fix each. Export blocked until resolved.
Jurisdiction mismatch	Cloned OrgDNA compliance pack does not cover new jurisdiction. System flags missing pack and suggests available alternatives.
Sensitive data warning	If any OrgApp has transactional data flagged for exclusion, system confirms exclusion list before export.

UC-04 Invite a User and Assign a Role [Settings / Auth]

Actor	Admin / Owner
Goal	Add a new team member to the OrgOS with the correct permissions.
Pre-conditions	Inviting user has Admin or Owner role. Invitee has an email address.
Post-conditions	New user is active in OrgOS with assigned role and ODD. Welcome email sent with getting-started guide.

Main Flow

1	Admin opens Settings → Users & Roles.
2	Admin clicks "+ Invite User". Enters invitee email address.
3	Admin selects a role from the role list or creates a new custom role.

4	Admin optionally selects OrgExec type (Human / AI Agent / Hybrid).
5	System sends invitation email with time-limited join link (48 hours).
6	Invitee clicks link. If they have no UME account, signup flow runs. If they do, they accept the invitation.
7	System adds the user to the OrgOS with the assigned role and generates their ODD from the role template.
8	Activity log entry created: user added, role assigned, ODD version recorded.

Alternate / Exception Flows

Invitation expires	After 48 hours, link is invalid. Admin can resend from the Invitations tab.
Custom role needed	Admin clicks "+ Create Role" and defines per-module permissions before sending invite.
Invitee already a member	System detects existing membership. Prompts admin to update role instead.

Group B — Finance & Accounting

UC-05 Create and Post a Journal Entry [Finance]

Actor	Accountant / Finance Manager
Goal	Record a financial transaction in the general ledger with a fully balanced, auditable journal entry.
Pre-conditions	User has Finance Write permission. Accounting period is open.
Post-conditions	JE posted to GL. Audit record with hash chain entry created. Period balance updated. All reports reflect new posting in real time.

Main Flow

1	User opens Finance → Journal Entries and clicks "+ New JE".
2	User enters entry date, selects accounting period, and adds a description.
3	User adds line items: selects account from COA autocomplete, enters debit or credit amount.
4	System calculates running balance after each line. BALANCED indicator shows green when debits equal credits.
5	User attaches a supporting document (receipt, invoice PDF).
6	User clicks "Post Entry". System validates: period open, entry balanced, no restricted accounts.
7	System generates SHA-256 hash, writes immutable audit record, updates GL balances, and emits finance.journal_entry.posted event.
8	SDO engine receives event; updates financial state in digital twin.

Alternate / Exception Flows

Period closed	If selected period is closed, system blocks posting and shows who closed the period and when.
Unbalanced entry	POST button is disabled while debits ≠ credits. Red unbalanced indicator shows the difference amount.
Requires approval	If entry amount exceeds the user's ODD approval threshold, system routes to approver instead of posting directly.
Duplicate detection	System warns if a near-identical entry (same date, amount, description) exists within 5 days.

UC-06 Generate and Send a Sales Invoice [Finance / CRM]

Actor	Owner / Sales / Finance
Goal	Create an invoice for a customer, send it, and track payment status.
Pre-conditions	Customer exists in CRM. User has Finance or Sales Write permission.
Post-conditions	Invoice sent. AR balance updated. On payment: JE posted, bank reconciliation item created, CRM order marked paid.

Main Flow

1	User opens Finance → Invoices and clicks "+ New Invoice" (or opens a CRM order and clicks "Generate Invoice").
2	System pre-populates customer details from CRM. User adds line items with descriptions and amounts.
3	System calculates subtotal, applies configured tax rate, shows total due.
4	User sets payment due date and selects payment method (bank transfer, Stripe link).
5	User previews invoice. Clicks "Send". System emails invoice to customer with payment link.
6	Invoice status changes to Sent. Dashboard AR widget updates.
7	On payment: Stripe webhook fires → system marks invoice Paid → posts receipts journal entry automatically → AR balance updated.

Alternate / Exception Flows

Customer pays partially	Partial payment recorded. Invoice status shows Partially Paid with outstanding balance.
Invoice overdue	At due date+1: status changes to Overdue. Dashboard risk widget shows AR ageing. Optional auto-reminder sent.
Credit note required	User clicks "Issue Credit Note" on a posted invoice. Reversal JE created automatically.

UC-07 Run Payroll for a Period [HR / Finance]

Actor	HR Manager / Payroll Administrator
Goal	Calculate and process payroll for all active employees, generate payslips, post payroll journals.
Pre-conditions	Employees have active payroll accounts. Pay period is defined. Timesheets/salaries are confirmed.
Post-conditions	Payroll posted. Payslips generated and accessible. Tax liability in compliance calendar. Finance GL updated.

Main Flow

1	HR Manager opens HR → Payroll → Run Payroll for the current period.
2	System pre-populates all active employees with their contracted salary or hours × rate from timesheets.
3	Manager reviews each employee row: gross pay, statutory deductions (tax, NI/FICA), net pay.
4	Manager enters any adjustments: overtime, bonuses, deductions.
5	System recalculates totals. Manager confirms payroll run.
6	System generates payslips for all employees, posts payroll journal entry to Finance (Gross Pay Dr, Tax Liability Cr, Net Pay Cr).
7	System marks payroll period as Processed. Employees notified by email. Payslips visible in employee portal.
8	Payroll tax liability entry appears in Chombo filing calendar as upcoming obligation.

Alternate / Exception Flows

Timesheet not submitted	System flags employees with missing timesheets. Manager can override with contracted hours or hold those employees.
Tax table update needed	System checks for jurisdiction tax table updates before each payroll run; warns if tables are stale.
Payroll ODD constraint	If payroll processor OrgExec's ODD requires dual-sign for amounts above threshold, approval request generated.

UC-08 View Consolidated Group P&L [Finance / DAO]

Actor	CFO / Group Finance Director
Goal	View a real-time consolidated profit and loss statement across all group entities.
Pre-conditions	User has Finance Read on all entities in the group. At least two entity OrgOS instances exist in a group DAO.
Post-conditions	Consolidated P&L displayed. All entity data reflected within SDO event propagation lag (< 500ms typical).

Main Flow

1	User opens Finance → Reports → P&L Statement.
2	User selects "All Entities" in the entity selector.
3	System aggregates revenue and expense data from all entity OrgOS instances via DAO data sharing.
4	System applies intercompany elimination rules (configured in group Finance settings).
5	Consolidated P&L renders with entity breakdown columns and group totals.
6	User can drill into any account row to see the entity-by-entity breakdown.

7	User exports to Excel or PDF for board pack.

Alternate / Exception Flows

Entity data stale	If a member entity's OrgOS is offline, system shows last-synced timestamp with staleness warning and excludes from totals.
Currency conversion	If entities report in different currencies, system applies configured exchange rates and shows functional and presentation currency.
Period not closed for entity	System flags entities with open periods and shows unaudited indicator on those columns.

UC-09 Reconcile a Bank Account [Finance]

Actor	Accountant / Finance Manager
Goal	Match bank statement transactions to GL entries, identify discrepancies, and mark the account as reconciled.
Pre-conditions	Bank feed connected via Finance driver or CSV import available. Unreconciled transactions exist.
Post-conditions	Bank account reconciled. Closing balance matches bank statement. Reconciliation report available for audit.

Main Flow

1	User opens Finance → Bank Reconciliation, selects account.
2	System shows two columns: GL transactions (unreconciled) and bank statement lines.
3	User matches GL transactions to bank lines by clicking pairs; matched items grey out.
4	System auto-suggests matches where amount, date, and payee match within configured tolerance.
5	User investigates unmatched items: posts adjusting JEs for bank charges, interest, errors.
6	When all items matched, user clicks "Mark Reconciled". System locks reconciliation and records closing balance.

Alternate / Exception Flows

Unmatched at period close	Outstanding unmatched items shown in summary. User can mark as "in transit" or raise outstanding item for investigation.
Duplicate bank line	System flags potential duplicate if same amount and payee appear within 2 business days.

UC-10 Close an Accounting Period [Finance]

Actor	Finance Manager / Controller
Goal	Lock an accounting period to prevent further posting and produce period-end reports.
Pre-conditions	All journal entries for the period are posted and reviewed. User has Finance Admin permission.
Post-conditions	Period locked. Period-end reports generated and immutable snapshot stored. DAO royalty smart contracts triggered if applicable.

Main Flow

1	User opens Finance → Period Management, selects the period to close.
2	System runs pre-close checklist: unreconciled bank accounts, unposted entries, unmatched invoices.
3	System shows any open items as warnings or blockers.
4	User resolves blockers (or acknowledges warnings with reason).
5	User clicks "Close Period". System prompts for confirmation.
6	System locks the period: all posting to this period is now blocked.
7	System generates period-end reports: P&L, Balance Sheet, Cash Flow statement.

8	Emits finance.period.closed event — subscribed by DAO smart contract engine for royalty triggers.

Alternate / Exception Flows

Open items block close	System lists each unresolved item. Finance Manager must either resolve or explicitly override with a reason logged in audit trail.
Reopening a period	Only Finance Admin can reopen. Requires dual authorisation if ODD requires. All reopen actions fully audited.

Group C — HR & People

UC-11 Onboard a New Employee (Full Wizard) [HR / ODD]

Actor	HR Manager
Goal	Create a complete employee record, generate their ODD contract, trigger cross-module provisioning events.
Pre-conditions	User has HR Write permission. Role and department exist in org structure.
Post-conditions	Employee record active. ODD registered. Cross-module cascade complete: payroll, contract, IT, portal access all provisioned. Timeline entry created.

Main Flow

1	HR Manager opens HR → People → "+ New Hire".
2	Step 1: Enters personal details, role, department, start date, employment type, reports-to. Selects OrgExec type (Human). Selects ODD template.
3	Step 2: Uploads right-to-work documents. System validates document type for jurisdiction.
4	Step 3: Enters compensation details: salary/rate, pay frequency, payroll account, equity (if applicable).
5	Step 4: Selects benefits package: health plan, pension/401k contribution.
6	Step 5: Review screen shows all details. System previews generated ODD contract. Manager confirms.
7	System creates employee record, registers OrgExec with generated ODD in ODDBus, triggers cross-module events.
8	Cross-module cascade: Finance creates payroll account; Legal generates employment contract; IT sends provisioning request; Portal sends welcome email with login.

Alternate / Exception Flows

Right-to-work fails	Document does not satisfy jurisdiction requirements. System flags specific gap. HR must resolve before proceeding.
Duplicate employee	System detects same name + DOB combination. Prompts to confirm this is a new person, not a duplicate.
ODD template not available	If no template matches the role, HR Manager prompted to create a custom ODD before proceeding.

UC-12 Process an Employee Leave Request [HR / Portal]

Actor	Employee → Manager
Goal	Employee submits a leave request; manager approves or declines; leave balance and schedule updated.
Pre-conditions	Employee has active portal access. Leave types and balances configured.
Post-conditions	Leave approved/declined. Balance updated. Schedule updated. Payroll informed. All actions audited.

Main Flow

1	Employee opens Employee Portal → Leave → "Request Leave".
2	Employee selects leave type, date range, and optional reason. System shows current balance and balance after approval.
3	Employee submits. System validates: sufficient balance, no scheduling conflict (via Operations module if connected).
4	System notifies manager (email + in-app alert).
5	Manager opens HR → Leave Inbox. Reviews request alongside team leave calendar.
6	Manager approves or declines with optional comment.
7	System updates leave balance, updates schedule in Operations module, notifies employee by email.
8	Leave calendar entry created. Payroll module informed for pay period calculation.

Alternate / Exception Flows

Insufficient balance	System blocks submission and shows available balance. Employee can request without-pay leave as alternative.
Manager unavailable	If manager has not actioned within configured SLA (default 3 days), system escalates to manager's manager.
Clash with critical period	If date range overlaps a flagged critical period (e.g., year-end), system warns manager during review.

UC-13 Deploy an AI Agent OrgExec [ODD / OrgExec]

Actor	Admin / AI Agent Manager / Owner
Goal	Configure, ODD-govern, and deploy an AI agent as an active OrgExec within the OrgOS.
Pre-conditions	User has OrgExec Admin permission. AI Agent OrgApp is installed and active.
Post-conditions	AI agent active as OrgExec. ODD registered and enforced. All agent actions logged in audit chain. SDO reflects new node.

Main Flow

1	User opens OrgSystem → OrgExec Registry → "+ New OrgExec".
2	User selects Type: AI Agent. Enters display name and role description.
3	User selects or creates an ODD template. ODD Editor opens.
4	User defines Capabilities: checks data sources the agent may read, outputs it may write, notifications it may send.
5	User defines Constraints: rate limits, working hours, approval thresholds (changes above N require human sign-off).
6	User defines Escalation Rules: conditions under which agent must pause and notify a human.
7	User runs "Test Agent" — system runs agent in sandbox against sample data.
8	Test passes. User clicks "Deploy OrgExec". System registers ODD in ODDBus, activates agent.
9	Agent appears in OrgExec Registry as Active. SDO topology graph updates.

Alternate / Exception Flows

Test fails	Agent cannot satisfy a required task within its ODD constraints. User revises capabilities or constraints and retests.
ODD too permissive	System flags if agent's ODD grants access to sensitive data (payroll, financial) without an explicit audit rule. Prompts to add audit constraint.
Agent hits rate limit	During operation, agent exceeds hourly execution limit. ODDBus suspends agent actions until next hour window. Alert sent to manager.

UC-14 Conduct a Performance Review [HR]

Actor	Manager → Employee
Goal	Initiate, complete, and record a structured performance review cycle.
Pre-conditions	Review cycle configured. Employee has active profile with performance targets set.
Post-conditions	Review completed and recorded. HR analytics updated with rating data. Compensation review trigger event emitted if configured.

Main Flow

1	Manager opens HR → Performance → Initiate Review for an employee.
2	System generates review form from configured template (goals, competencies, ratings).
3	Employee receives portal notification to complete self-assessment.
4	Employee submits self-assessment. Manager is notified.
5	Manager completes manager assessment with ratings and narrative.

6	Optionally: 360 feedback requested from peers/direct reports.
7	Manager schedules and holds review meeting. Meeting notes recorded in system.
8	Manager submits completed review. Review is shared with employee via portal.
9	Employee acknowledges receipt. Record marked Complete. HR can view aggregate ratings for team.

Alternate / Exception Flows

Employee does not complete self-assessment	System sends reminder at T+3 days. Manager can override and complete without self-assessment if needed.
Rating dispute	Employee flags disagreement on portal. HR notified. Escalation workflow opens.

UC-15 Offboard an Employee [HR / Legal / IT]

Actor	HR Manager
Goal	Process an employee's departure: close access, complete final pay, archive records.
Pre-conditions	Employee is active. Offboarding date confirmed. HR Admin permission required.
Post-conditions	Employee status Inactive. ODD suspended. Access revoked. Final payslip generated. Legal documents archived. IT deprovisioned.

Main Flow

1	HR Manager opens employee profile → "Offboard Employee".
2	Offboarding wizard opens. HR enters last day, reason for leaving (voluntary/involuntary), notice period status.
3	System generates offboarding checklist: exit interview, equipment return, access revocation, final payslip.
4	HR works through checklist items. Each item ticked off creates a timeline entry.
5	On last day: system automatically revokes OrgOS access, suspends ODD contract, archives employee record.
6	Finance generates final payslip including any outstanding holiday pay/severance.
7	Legal module generates separation agreement from template if applicable.
8	IT provisioning event emitted: connected systems (email, Slack) deprovisioned via device drivers.

Alternate / Exception Flows

Equipment not returned	Outstanding equipment tracked in IT module. Flag remains on HR record until cleared.
Garden leave period	Access revoked immediately but payroll continues until end of notice period.

UC-16 Register an OrgCPU (IoT Device) [ODD / Operations / IT]

Actor	IT Manager / Facilities Manager
Goal	Connect a physical device (machine, POS terminal, sensor) as an OrgCPU and define its ODD.
Pre-conditions	Device is network-accessible. IT Admin permission held.
Post-conditions	OrgCPU registered and active. ODD enforced. Live data flowing to configured OrgApps. SDO topology updated.

Main Flow

1	User opens OrgSystem → OrgExec Registry → "+ New OrgExec" → Type: OrgCPU.
2	User selects device category (Industrial Machine / POS Terminal / IoT Sensor / 3rd-party Platform).
3	User enters device name, physical location, and connection details (IP, API endpoint, MQTT topic).
4	System runs connectivity test. Confirms device is reachable.
5	User selects ODD template for the device type (e.g., IOT-WM-1 for industrial washer).
6	User configures: what data the device reports, at what frequency, and which OrgApps receive the data.
7	User defines alert thresholds (e.g., temperature > 90°C → alert; status = offline > 5 min → alert).

8	User deploys ODD. System registers device in OrgCPU Registry and SDO topology.
9	Device status appears live in Operations dashboard.

Alternate / Exception Flows

Connectivity test fails	User provided wrong endpoint. System shows error detail. User corrects and retests.
No ODD template for device	User creates custom ODD with IT Manager guidance. Saved to OrgDNA library for future use.

Group D — Legal & Compliance

UC-17 Register a Legal Entity [Legal / Chombo]

Actor	Owner / Legal/Compliance Officer
Goal	Add a new legal entity to the OrgOS, set jurisdiction, and auto-generate compliance obligations.
Pre-conditions	User has Legal Admin permission.
Post-conditions	Entity registered. Compliance calendar populated. Entity linked to Finance and HR modules. Dashboard shows entity's first upcoming obligation.

Main Flow

1	User opens Legal & Compliance → Entities → "+ New Entity".
2	User enters entity name, entity type (LLC, Ltd, Corporation, etc.), registration number, registered address.
3	User selects primary jurisdiction. System detects applicable compliance framework.
4	System auto-generates compliance calendar: all statutory filings, annual returns, tax deadlines for this entity and jurisdiction.
5	User reviews generated obligations. Can add custom obligations or suppress irrelevant ones.
6	User links entity to OrgApps (Finance, HR) for cross-module data association.
7	Entity saved. Compliance calendar active. Dashboard compliance widget updated.

Alternate / Exception Flows

Jurisdiction not in database	System flags limited coverage. User can manually add compliance obligations. System flags entity as "manual calendar"
Entity type changes	If entity converts (e.g., sole trader to LLC), user runs "Convert Entity" wizard. Old obligations archived, new ones generated.

UC-18 Mark a Compliance Filing as Complete [Legal / Chombo]

Actor	Owner / Compliance Officer
Goal	Record that a compliance obligation has been fulfilled, upload evidence, and schedule next occurrence.
Pre-conditions	Filing obligation exists in compliance calendar. Due date is approaching or passed.
Post-conditions	Obligation marked Complete with evidence. Next occurrence scheduled. Compliance score updated. Audit record created.

Main Flow

1	User opens Legal → Filing Calendar. Sees filing due (highlighted amber or red).
2	User clicks on the filing. Detail screen shows: obligation description, instructions, due date, linked amount (if applicable, e.g., tax payment auto-calculated from payroll data).
3	User completes the filing with the relevant authority (outside UME).
4	User returns to UME. Clicks "Mark as Filed". Enters filing date and confirmation reference.
5	User uploads evidence document (EFTPS confirmation, Companies House receipt, etc.).
6	System marks obligation Complete, records filing date and reference, stores evidence in document vault.
7	System auto-creates next occurrence of recurring obligation (e.g., next quarterly return).
8	Compliance calendar and dashboard compliance score updated.

Alternate / Exception Flows

Filing late	System records late filing flag. Compliance score penalised. User can note reason for lateness.
Delegate to team member	User clicks "Assign To" before marking filed. Assignee receives task notification.

UC-19 Generate a Contract from Template [Legal / Chombo]

Actor	Owner / Legal Officer / HR Manager
Goal	Produce a ready-to-sign contract document using a configured template and party-specific data.
Pre-conditions	Contract template exists in Legal vault. Relevant party data exists (employee record or customer record).
Post-conditions	Contract executed and stored. legal.contract.executed event emitted. Document vault entry created. Cross-module cascade: HR links to employee record, Finance links to billing schedule.

Main Flow

1	User opens Legal → Contracts → "+ Generate Contract".
2	User selects contract type (Employment Agreement, NDA, Services Agreement, Partnership Agreement, Franchise Agreement).
3	System loads template. User selects counterparty from CRM or HR records.
4	System pre-populates template fields from OrgOS data (org name, address, entity details, party name, role, salary).
5	User reviews and edits the populated contract. Adds any bespoke clauses.
6	User sends for e-signature via integrated e-sign driver. Counterparty receives email with signing link.
7	On all parties signing: system marks contract Executed, stores signed PDF in document vault, emits legal.contract.executed event.

Alternate / Exception Flows

Counterparty declines	System records decline. Contract moves to Declined status. Notification sent to initiating user.
Template missing a field	User can fill manually. Field flagged as manually populated in audit record.
Jurisdiction-specific clauses needed	Legal template library includes jurisdiction variants. User selects applicable variant before generating.

UC-20 Manage IP, Licences & Brand Assets [Legal / Brand]

Actor	Owner / Brand Manager / Legal Officer
Goal	Register intellectual property, trademarks, and brand assets; track renewal dates.
Pre-conditions	User has Legal Write permission.
Post-conditions	IP asset registered. Renewal in compliance calendar. Brand assets accessible to Marketing module.

Main Flow

1	User opens Legal → IP Management → "+ Register Asset".
2	User selects asset type: Trademark, Patent, Copyright, Trade Secret, Domain Name, Brand Asset.
3	User enters registration details: title, jurisdiction, registration number, registration date, expiry/renewal date, owner entity.
4	User uploads supporting documents (registration certificate, design files).
5	System saves asset record. Adds renewal date to compliance calendar with configured reminder lead time.
6	Brand assets (logos, style guides) stored in Brand Assets sub-module; accessible to Marketing OrgApp.

Alternate / Exception Flows

Renewal approaching	Compliance calendar fires reminder 60 days before expiry. Legal officer assigns renewal task.
Cross-jurisdiction registration	User registers same IP in multiple jurisdictions; system groups under parent IP record with per-jurisdiction entries.

UC-21 Produce a Tax Audit Data Pack [Legal / Finance]

Actor	Owner / Finance Manager / Legal Officer
Goal	Prepare and export a complete, structured data pack for a tax authority audit.
Pre-conditions	Finance and Legal modules have data for the audit period. User has Finance and Legal Admin permission.
Post-conditions	Audit pack generated. All relevant documents linked and exported. Auditor access provisioned if requested.

Main Flow

1	User opens Legal → Compliance → "Prepare Audit Pack".
2	User selects audit scope: entity, tax type (income, sales, payroll), period range.
3	System compiles: all JEs for period, payroll records, filed returns with evidence, vendor invoices, employee contracts.
4	User reviews the compiled pack. Adds a cover note with organisation details and legal representative.
5	User generates audit pack (PDF or ZIP with structured folder organisation).
6	User grants auditor a time-limited read-only access token to the UME audit dashboard, or downloads pack for delivery.

Alternate / Exception Flows

Auditor needs direct system access	User creates a temporary read-only auditor OrgExec with AUDITOR-ODD (read-only, scoped to audit period, auto-expiring).
Missing documents	System flags JEs with no attached receipt. User can annotate with explanation or upload late documents.

Group E — Sales & CRM

UC-22 Create and Manage a Customer Order [CRM / Finance]

Actor	Front Desk / Sales Staff
Goal	Record a new customer order, track it through fulfilment, and mark it complete.
Pre-conditions	Customer exists in CRM (or is created). Sales module active.
Post-conditions	Order fulfilled, collected, and invoiced. Revenue posted to Finance. Customer record updated with order history.

Main Flow

1	Staff opens Sales → Order Board → "+ New Order" (or opens customer profile → "+ New Order").
2	Staff enters items/services, quantities, special instructions, and due date.
3	System calculates price from configured price list. Applies any customer-specific discount.
4	Order saved with status "Received". Card appears in Received column of order kanban.
5	Staff or OrgCPU moves order through kanban stages (Received → In Progress → Ready → Collected) as work proceeds.
6	On status change to "Ready": system sends customer notification (SMS/email) from configured template.
7	Staff confirms customer collection. Order marked Collected. Invoice generated automatically.
8	Finance module records revenue. AR balance updated.

Alternate / Exception Flows

Customer disputes item	Staff opens order detail → Add Dispute Note. Manager notified. Order held until resolved.
Order overdue	If order reaches due date without moving to Ready, system flags amber on kanban card. Manager alerted.
Bulk orders	Multi-item orders tracked with per-item status so partial readiness is visible.

UC-23 Manage a Sales Pipeline (B2B) [CRM / Sales]

Actor	Sales Manager / Business Development
Goal	Track B2B deals from lead to close using the CRM pipeline.
Pre-conditions	CRM module active. Sales pipeline stages configured.
Post-conditions	Deal won or lost. Revenue forecast updated. Finance notified. Win/loss analytics updated.

Main Flow

1	Sales rep opens CRM → Pipeline → "+ New Deal".
2	Enters deal name, associated company/contact, estimated value, expected close date, and initial stage.
3	Deal card appears on pipeline kanban board in the configured first stage.
4	Rep progresses deal: logs calls, emails, and meetings as activities against the deal record.
5	Rep moves deal through stages as milestones are achieved (Qualified → Proposal → Negotiation → Won/Lost).
6	On Won: system creates a sales order or project record. Finance notified for invoicing.
7	On Lost: system captures loss reason. Analytics module updates win-rate KPIs.
8	Pipeline value and stage distribution visible in Sales Dashboard.

Alternate / Exception Flows

Deal goes cold	Rep marks deal as On Hold. Deal greys on kanban. System sends re-engagement reminder after configured days.
Multi-stakeholder deal	Multiple contacts linked to a deal. Activity log tracks all interactions. Deal requires sign-off from two contacts.

Forecast override	Manager can override AI probability forecast with manual assessment; both values tracked.

UC-24 Send a Marketing Campaign to a Customer Segment [Marketing / CRM]

Actor	Marketing Manager / Owner
Goal	Create a targeted email or social campaign using a micro-persona, schedule it, and track engagement.
Pre-conditions	Marketing OrgApp active. Customer segment defined in CRM. Email/social channel driver configured.
Post-conditions	Campaign sent. Engagement tracked. Revenue attribution recorded. Customer segment updated with campaign interaction history.

Main Flow

1	User opens Marketing → Campaigns → "+ New Campaign".
2	Selects campaign type (email, social, SMS). Enters campaign name and goal.
3	User selects target segment from CRM (e.g., "Customers not seen in 60+ days").
4	User opens AI Campaign Generator, selects micro-persona and brand tone. Clicks "Generate Content".
5	AI generates subject line, body copy, and social post variants. User reviews and edits.
6	User schedules campaign send date/time.
7	System sends campaign via configured CommDriver. Opens, clicks, and conversions tracked.
8	Campaign results appear in Marketing Dashboard: open rate, click rate, revenue attributed.

Alternate / Exception Flows

Email bounce rate high	System flags high bounce rate after first batch. User reviews and cleans segment before continuing.
GDPR/CAN-SPAM compliance	System checks all recipients have opted-in to marketing. Suppresses non-consented contacts automatically.
A/B test	User creates two content variants. System splits segment 50/50. Winner deployed after configured test duration.

UC-25 Manage Customer Accounts Receivable [Finance / CRM]

Actor	Finance / Sales
Goal	Monitor outstanding invoices, send reminders, and record payments.
Pre-conditions	Invoices exist in Finance with Sent or Overdue status.
Post-conditions	Outstanding invoices actioned. Payments recorded. AR ageing report current. Cash flow forecast updated.

Main Flow

1	User opens Finance → AR Ageing report.
2	Report shows all outstanding invoices grouped by ageing bucket: 0–30, 31–60, 61–90, 90+ days.
3	User filters to overdue invoices. Selects one.
4	User clicks "Send Reminder" — system sends a payment reminder email using AR reminder template.
5	Customer pays via bank or Stripe payment link.
6	Payment recorded (automatically via Stripe webhook or manually by user).
7	Invoice marked Paid. AR balance updated. Cash reconciliation item created.

Alternate / Exception Flows

Customer queries invoice	User adds a note to invoice record. Customer portal shows note. Finance can issue credit note if error confirmed.
Debt referred to collections	User marks invoice "In Collections". Finance driver can route to third-party collections platform.

UC-26 Build and Publish a Customer Micro-Persona [Marketing]

Actor	Marketing Manager / Owner
Goal	Define a target customer segment persona for use in AI campaign generation.
Pre-conditions	CRM has customer data (optional but enriches persona). Marketing module active.
Post-conditions	Persona saved and available for campaign targeting.

Main Flow

1	User opens Marketing → Micro-Personas → "+ New Persona".
2	User enters persona name and description.
3	User selects demographic attributes: age range, location, income bracket.
4	User enters behavioural attributes: primary needs, pain points, buying triggers, preferred channels.
5	Optionally: user clicks "Generate from CRM data" — AI analyses existing customer records to suggest attribute values.
6	User reviews and adjusts AI suggestions. Saves persona.
7	Persona is now available as a target in the AI Campaign Generator and segment builder.

Alternate / Exception Flows

Data not available	User manually enters all attributes. AI generation requires at least 20 customer records for meaningful analysis.

Group F — Operations & Supply Chain

UC-27 Manage Inventory and Trigger a Purchase Order [Operations / Finance]

Actor	Operations Manager / Purchasing
Goal	Monitor stock levels, identify low-stock items, and create purchase orders to replenish.
Pre-conditions	Inventory items configured with reorder points. Vendor records exist.
Post-conditions	Inventory replenished. PO fully received and closed. Vendor AP invoice matched to PO receipt.

Main Flow

1	Operations Manager opens Operations → Inventory Dashboard.
2	Low stock alerts shown: items at or below reorder point highlighted in red.
3	Manager clicks "[Create PO]" on a low-stock item. New PO form opens, pre-populated with item and recommended quantity.
4	Manager selects vendor (preferred vendor auto-selected if configured), sets delivery date.
5	Manager reviews and submits PO. Status: Submitted.
6	Vendor notification sent via CommDriver (email or EDI).
7	Vendor confirms. PO status → Confirmed.
8	Goods arrive. Manager opens PO → "Receive Goods". Enters actual quantities received. Discrepancies flagged.
9	Stock levels updated. Finance records goods receipt for AP matching.

Alternate / Exception Flows

Auto-reorder enabled	If auto-reorder configured for item, system creates PO draft automatically when reorder point breached. Manager approves draft.
Vendor out of stock	Manager selects alternative vendor from vendor list. Original vendor backorder noted on PO.
Short delivery	Partial goods receipt recorded. PO remains open for outstanding quantity. Finance AP updated for partial amount.

UC-28 Monitor OrgCPU Equipment Health [Operations / ODD]

Actor	Operations Manager / Facilities Manager
Goal	View live status of connected machinery, respond to alerts, and log maintenance.
Pre-conditions	OrgCPU devices registered and active with IoT ODDs.
Post-conditions	Equipment health current. Maintenance logged. Cost posted. Downtime tracked for KRI reporting.

Main Flow

1	Manager opens Operations → Equipment Dashboard.
2	All registered OrgCPUs shown with live status (Running / Idle / Alert / Offline) and health score.
3	Manager clicks on a machine to view detail: current readings, last 24h history, next maintenance date.
4	On alert: manager sees alert description and recommended action.
5	Manager assigns maintenance task to a technician OrgExec via Work module.
6	Technician completes maintenance, logs completion and parts used in maintenance log.
7	Machine status updated. Next maintenance date advanced. Maintenance cost posted to Finance.

Alternate / Exception Flows

Machine offline	System escalates offline alert after 5 minutes. Risk module KRI "Equipment Uptime" updated. Dashboard shows critical flag.

ODD reading anomaly	ODD constraint on reading value triggers alert if outside normal range. AI OrgExec can be configured to diagnose pattern.

UC-29 Process a Supplier Invoice (AP) [Finance / Operations]

Actor	Finance / Accounts Payable
Goal	Record an incoming supplier invoice, match to PO, approve, and schedule for payment.
Pre-conditions	Purchase order exists in Received status. Supplier invoice received.
Post-conditions	Supplier invoice paid. AP balance updated. GL posted. Bank reconciliation item created.

Main Flow

1	User opens Finance → Accounts Payable → "+ New Supplier Invoice".
2	User enters supplier, invoice number, date, and line items.
3	System suggests PO matches based on supplier and amount. User confirms match.
4	Three-way match validated: PO quantity, goods receipt quantity, invoice quantity must align.
5	If match: invoice auto-approved within configured tolerance. If discrepancy: held for manual review.
6	Approved invoice scheduled for payment on due date. Payment run processes on schedule via Finance driver.
7	Payment confirmed. Invoice marked Paid. Audit record created.

Alternate / Exception Flows

Three-way match fails	User reviews discrepancy report. Raises supplier query or adjusts goods receipt. Manual approval required from Finance Manager.
Early payment discount	System detects early payment discount term. Shows net amount if paid within discount window. User elects to accelerate.

Group G — Risk & GRC

UC-32 Create and Maintain a Risk Register Entry [Risk / GRC]

Actor	Risk Manager / Owner
Goal	Add a risk, assess it, assign an owner, and track mitigations.
Pre-conditions	Risk & GRC module active. User has GRC Write permission.
Post-conditions	Risk registered. Owner assigned. Heat map updated. KRI linked. Executive dashboard risk widget updated.

Main Flow

1	User opens Risk → Register → "+ New Risk".
2	User enters risk title, description, category, owner, and source (operational/financial/compliance/strategic).
3	User assesses inherent risk: selects Likelihood (1–5) and Impact (1–5). System calculates inherent risk score.
4	User enters mitigation actions (existing controls). Re-assesses residual likelihood and impact.
5	System calculates residual risk score. Plots on heat map.
6	Risk assigned to owner. Owner receives notification to confirm ownership.
7	System links risk to relevant KRI if available (e.g., cash runway KRI linked to "liquidity risk").
8	Risk visible in register and heat map dashboard.

Alternate / Exception Flows

Risk escalates	If residual risk score exceeds threshold, system auto-escalates to board-level risk register and notifies exec.
KRI breach	Linked KRI breach auto-updates risk status to "Materialising". Incident log entry created.

UC-33 Respond to a KRI Threshold Breach [Risk / GRC]

Actor	System → Executive / Risk Manager
Goal	Detect a KRI breach, alert the responsible executive, and record the response.
Pre-conditions	KRIs configured with thresholds. KRI monitoring running.
Post-conditions	Breach acknowledged. Incident created. Response underway. All actions audited. SDO risk layer updated.

Main Flow

1	System evaluates KRI against current data from linked OrgApp (Finance cash balance, HR utilisation, etc.).
2	KRI breaches threshold. System creates KRI breach record with timestamp, current value, and threshold value.
3	System emits risk.kri.breach event. Dashboard shows red breach card with sparkline trend.
4	Risk module notifies responsible OrgExec (from KRI configuration).
5	Responsible exec opens KRI dashboard, reviews breach detail.
6	Exec clicks "Acknowledge" — records that exec has seen the breach.
7	Exec clicks "Escalate" or "Create Incident" — routes to Risk Incident log for formal response tracking.
8	Response actions assigned. Progress tracked. KRI value monitored for recovery.

Alternate / Exception Flows

KRI recovers before action	If KRI value returns to within threshold before response is completed, system marks as "Auto-recovered" but keeps incident open for review.
False trigger	Exec marks breach as "False Positive" with reason. Threshold recalibration recommended.

UC-34 Run a GRC Compliance Assessment [Risk / GRC]

Actor	Compliance Officer / Risk Manager
Goal	Evaluate the organisation's compliance posture against a framework (e.g., SOC 2, ISO 27001).
Pre-conditions	GRC framework configured. User has GRC Write permission.
Post-conditions	Assessment complete. Compliance score calculated. Gaps documented. Remediation tasks created. Report generated.

Main Flow

1	User opens Risk & GRC → Assessments → "+ New Assessment".
2	User selects compliance framework from library (SOC 2, ISO 27001, GDPR, local AML, etc.).
3	System generates assessment questionnaire from framework control set.
4	User works through controls: marks each as Compliant / Partially Compliant / Non-Compliant / Not Applicable.
5	For each gap: user enters remediation plan, owner, and target date.
6	System calculates overall compliance score and generates gap summary.
7	Assessment report generated (PDF). Can be shared with auditors or board.
8	Remediation tasks created in Work module. Progress tracked to next assessment.

Alternate / Exception Flows

Control evidence required	User attaches evidence documents to each control assertion from the document vault.

UC-35 Manage a Board / Governance Meeting [Board / Legal]

Actor	Board Secretary / Executive
Goal	Schedule a board meeting, distribute papers, record minutes, and track resolutions.
Pre-conditions	Board Management module active. Board members registered as OrgExecs with BOARD-ODD.
Post-conditions	Meeting recorded. Minutes approved and stored. Resolutions in register. Action items assigned.

Main Flow

1	Secretary opens Board Management → Meetings → "+ New Meeting".
2	Enters meeting date, type (board/committee/AGM), location/video link, and agenda items.
3	System notifies board members. Members confirm attendance.
4	Secretary uploads board papers to meeting record. Members access via portal.
5	Meeting occurs. Secretary records attendance, discussions, and resolutions in minutes template.
6	Draft minutes circulated to board members for review.
7	Board members approve minutes (e-signature or portal acknowledgement).
8	Approved minutes and resolutions stored in corporate secretariat vault. Resolution register updated.
9	Action items from resolutions assigned to OrgExecs via Work module.

Alternate / Exception Flows

Quorum not met	System flags quorum shortfall. Meeting may proceed for discussion only; resolutions deferred until quorum achieved.
Urgent resolution	Emergency board consent process: resolution circulated by email; members sign individually; system records responses.

Group H — OrgSystem (ODD, SDO, OrgApps)

UC-36 Install an OrgApp from the Marketplace [OrgSystem / OrgApps]

Actor	Admin / Owner
Goal	Browse the OrgApp Marketplace, install a new app, and configure it for the OrgOS.
Pre-conditions	User has OrgApp Admin permission. OrgOS is running.
Post-conditions	OrgApp running. SDO topology updated. OrgDNA v+1 committed. App accessible to users with appropriate ODD permissions.

Main Flow

1	User opens OrgSystem → OrgApp Marketplace.
2	User browses by category or searches by name. Finds the desired OrgApp.
3	User clicks on app card. Detail page shows: description, permissions required, pricing tier, reviews, dependencies.
4	User clicks "Install". System checks tier eligibility and dependency availability.
5	Configuration wizard opens, driven by the OrgApp's config_schema(). User fills in required settings.
6	User grants requested permissions. Reviews capability grants that this app will make available to OrgExecs.
7	User clicks "Complete Setup". System starts the OrgApp, registers it in OrgKernel, updates SDO topology.
8	New OrgApp visible in OrgOS Control Panel as Running. OrgDNA version incremented.

Alternate / Exception Flows

Tier restriction	App requires Enterprise tier but org is on Growth. System shows upgrade path. Admin can trial for 14 days.
Dependency missing	App requires Finance module which is not active. System prompts to enable dependency first.
Config validation fails	Required config field left empty. Wizard blocks progress until resolved.

UC-37 View and Interact with the Digital Twin (SDO) [SDO / OrgSystem]

Actor	Executive / Owner / Digital Twin Analyst
Goal	Open the SDO view, understand the live organisational topology, and identify items needing attention.
Pre-conditions	SDO Engine active. At least 3 OrgApps running.
Post-conditions	SDO reviewed. Actions identified and initiated. Org health score visible.

Main Flow

1	User opens OrgSystem → Digital Twin (SDO).
2	SDO overview shows: live topology graph with all OrgApps, OrgExecs, and OrgCPUs as nodes; data flow edges between them; colour-coded health.
3	User clicks on a node (e.g., Finance OrgApp) to see detail: status, CPU%, last event, active records, linked OrgExecs.
4	User clicks "View Flow Map" for the interactive version with animated data flow.
5	SDO Health Score panel shows: score out of 100, breakdown by category (KRI breaches, ODD violations, OrgApp errors, compliance gaps).
6	User clicks "AI Optimisation Suggestions". OrgOptimiser analyses SDO state and returns top 3 recommendations.
7	User selects a recommendation, reviews detail, and launches the suggested action (e.g., adjust KRI threshold, re-schedule task, add OrgExec capacity).

Alternate / Exception Flows

Node offline	Offline node shown in grey. Click shows last known state and offline duration.

Optimiser suggestions not available	Insufficient data for meaningful suggestions (new OrgOS < 30 days old). System shows date when suggestions will be available.

UC-38 Run an OrgOS Simulation (What-If) [SDO / OrgSimulator]

Actor	Executive / Digital Twin Analyst / Strategist
Goal	Model the impact of a business decision on the organisation before committing to it.
Pre-conditions	SDO Engine active with OrgSimulator. Sufficient historical data exists.
Post-conditions	Simulation complete. Results exported. Decision informed by modelled impact.

Main Flow

1	User opens OrgSystem → Digital Twin → OrgSimulator → "+ New Simulation".
2	User selects simulation type: Headcount Change / Revenue Scenario / Cost Reduction / New Market Entry.
3	User sets scenario parameters (e.g., "Hire 3 staff in Q3; Revenue +15% from October").
4	User sets simulation horizon (3 months / 6 months / 12 months).
5	User clicks "Run Simulation". System applies scenario to current SDO state, projects forward using financial and operational models.
6	Results show: projected P&L impact, cash runway change, KRI impact, headcount cost, risk score change.
7	User adjusts parameters and re-runs. Compares scenarios side by side.
8	User exports simulation report for decision-making or board presentation.

Alternate / Exception Flows

Simulation exceeds model confidence	System flags projections beyond 6 months as lower confidence. Sensitivity ranges shown rather than point estimates.

UC-39 Configure OrgOS Roles and Permissions [Settings / ODD]

Actor	Admin / Owner
Goal	Define custom roles with granular per-module permissions and assign to users.
Pre-conditions	Admin permission held.
Post-conditions	Role created and assigned. Users gain new permissions. ODD templates linked and registered.

Main Flow

1	User opens Settings → Users & Roles → Roles tab → "+ Create Role".
2	User enters role name and description.
3	Permission matrix shown: rows = OrgApps, columns = access levels (None / Read / Write / Admin).
4	User sets permissions per module.
5	User optionally links role to an ODD template (so users assigned this role automatically get a matching ODD).
6	Role saved. User assigns role to team members from Users tab.

Alternate / Exception Flows

Conflicting permissions	If a user is assigned two roles, the more permissive of the two applies per module by default. Admin can set "least-privilege" mode.
Role used by AI agent	Role assigned to AI OrgExec: ODD enforcement applies on top of role permissions (ODD is the binding constraint).

UC-40 View the Full Audit Trail [Settings / Audit]

Actor	Auditor / Compliance Officer / Admin
Goal	Search and export the immutable audit chain for a specified scope and period.
Pre-conditions	Audit trail populated (all actions audited by default). User has Audit Read permission.
Post-conditions	Audit records retrieved, reviewed, and exported. Chain integrity verified.

Main Flow

1	User opens Settings → Audit Log (or Finance → Audit if scoped to Finance).
2	User sets filters: actor, action type, resource, date range, outcome (Success/Denied/Escalated).
3	System returns matching audit records from immutable append-only store.
4	User can view individual record detail: actor, action, resource, timestamp, IP, device, session ID, outcome.
5	User clicks "Verify Hash" on any record to confirm cryptographic chain integrity.
6	User exports filtered results as CSV or PDF for external audit.

Alternate / Exception Flows

Hash verification fails	System flags integrity breach alert. Admin notified immediately. Security incident workflow triggered.

UC-41 Upgrade an ODD Contract [ODD / OrgSystem]

Actor	Admin / HR Manager / AI Agent Manager
Goal	Update an OrgExec's ODD contract to a new version without interrupting their active work.
Pre-conditions	OrgExec has an active ODD. New ODD version prepared.
Post-conditions	ODD upgraded. Old version archived. New version active. Audit record created. OrgDNA updated.

Main Flow

1	User opens OrgSystem → OrgExec Registry → selects OrgExec.
2	User clicks "Edit ODD". ODD Editor opens showing current contract version.
3	User makes changes: updates capabilities, constraints, or data access grants.
4	User enters change reason. Clicks "Save as New Version".
5	System shows diff between old and new version. User confirms.
6	System applies graceful transition: in-flight actions complete under old contract.
7	New ODD version activated. Old version archived with transition timestamp.
8	All future actions validated against new ODD. Audit record: ODD upgraded, by whom, from/to versions.

Alternate / Exception Flows

In-flight escalation	An escalated action from old ODD is pending approval when ODD is updated. Escalation resolves under old ODD terms.
Downgrading capabilities	System warns if new ODD removes a capability currently in active use. User must confirm intent.

Group I — DAO & Distributed Ledger

UC-42 Form a DAO [DAO / DLT]

Actor	Franchise Owner / Group CEO / DAO Founder
Goal	Create a Decentralised Autonomous Organisation connecting multiple OrgOS instances under shared governance.

Pre-conditions	User has DAO Admin permission. At least two OrgOS instances are accessible. DAO module active.
Post-conditions	DAO live. DLT network active with consensus. Smart contracts deployed. All members' SDOs updated with DAO node connection.

Main Flow

1	User opens DAO & DLT → "+ Form New DAO".
2	User enters DAO name, description, and governance model (majority vote / supermajority / one-org-one-vote / token-weighted).
3	User sets quorum requirement and voting period.
4	User defines initial smart contract terms using the visual contract builder (royalties, brand standards, SLA terms).
5	User publishes current OrgDNA as the canonical DAO brand/base package.
6	User invites member organisations by entering their OrgOS endpoint or UME org ID.
7	Each invited org receives an invitation. Their admins review contract terms and multi-sign acceptance.
8	On all founding members accepting: DLT network initialised, genesis block committed, DAO goes live.
9	DAO dashboard shows member network, treasury, and ledger.

Alternate / Exception Flows

Member declines contract terms	Member org can propose amendments via governance channel before accepting. If unresolved, they remain non-member.
Consensus failure during genesis	If fewer than minimum members accept, DAO formation aborted and retry available.

UC-43 Accept an OrgDNA Update as a DAO Member [DAO / OrgDNA]

Actor	Franchisee Owner / Member Org Admin
Goal	Receive, review, and accept an OrgDNA update broadcast by the parent org.
Pre-conditions	Organisation is an active DAO member. An OrgDNA update has been broadcast.
Post-conditions	OrgDNA updated. DLT block committed. Brand standards compliance confirmed. OrgDNA version history updated.

Main Flow

1	System receives OrgDNA broadcast from parent org via DAO inter-org event bus.
2	Admin receives notification: "New OrgDNA update v2.1 available from [Parent Org]".
3	Admin opens DAO & DLT → OrgDNA Updates. Sees broadcast with version number and change summary.
4	Admin clicks "View Diff". System shows component-by-component changes: new compliance pack, updated COA entries, revised ODD templates.
5	Admin reviews impact. Can add to staging environment for testing before applying.
6	Admin clicks "Accept Update". System applies OrgDNA diff to this OrgOS. Validates. Activates changes.
7	ORGDNA_UPDATE transaction committed to DLT recording acceptance.
8	Brand standards smart contract check triggered: confirms this org is now compliant with parent's latest version.

Alternate / Exception Flows

Update introduces breaking change	System flags breaking change (e.g., COA restructure requiring data migration). Admin must complete migration wizard before accepting.
Mandatory update	If update is marked mandatory by parent with cure period, system shows deadline countdown. Non-acceptance after deadline triggers DAO smart contract warning.
Update not applicable	Admin can decline optional updates with reason. Recorded on DLT as acknowledged-not-accepted.

UC-44 Participate in a DAO Governance Vote [DAO / Governance]

Actor	DAO Member Admin / Owner
Goal	Vote on a governance proposal submitted by another DAO member.
Pre-conditions	Active DAO membership. A governance proposal is open for voting.
Post-conditions	Vote cast and recorded on DLT. Result published. Approved actions executed or assigned.

Main Flow

1	System sends notification: "New governance proposal: [Title] — voting open until [date]".
2	Member opens DAO & DLT → Governance → open proposal.
3	Member reviews proposal: description, rationale, proposed action, supporting documents.
4	Member can post a comment or question to the governance discussion thread.
5	Member casts vote: For / Against / Abstain.
6	Vote recorded on DLT. Running tally updates in real time (anonymised until voting closes if configured).
7	On voting period close: system tallies votes, checks quorum. If quorum met and threshold passed: proposal approved.
8	Approved action executed automatically if it is a smart contract execution (e.g., fee change). Manual actions assigned as tasks.

Alternate / Exception Flows

Quorum not met	Proposal enters extended voting period. If still no quorum after extension, proposal lapses.
Urgent vote	Emergency proposals have 24-hour window with >75% threshold required. All members notified by push + email immediately.

UC-45 View and Verify the DLT Ledger [DAO / DLT]

Actor	Auditor / DAO Member / Compliance Officer
Goal	Browse inter-organisation transactions on the distributed ledger and verify block integrity.
Pre-conditions	Active DAO membership. DLT has transactions recorded.
Post-conditions	Transactions reviewed. Hashes verified. Export produced.

Main Flow

1	User opens DAO & DLT → Distributed Ledger.
2	User sees transaction list: block number, timestamp, type, from org, to org, amount (if financial).
3	User filters by transaction type, organisation, or date range.
4	User clicks a transaction to see full detail: all fields, payload hash, signatories, consensus state.
5	User clicks "Verify Hash". System recomputes block hash and confirms it matches stored hash and chain linkage.
6	User exports selected transactions as CSV for external audit or reporting.

Alternate / Exception Flows

Hash mismatch	System flags ledger integrity alert. DAO governance notified. Security incident raised.
Transaction not yet confirmed	Unconfirmed transactions shown in mempool view with pending indicator.

UC-46 Execute a Smart Contract Royalty Payment [DAO / DLT / Finance]

Actor	System (automatic)
Goal	Automatically transfer a royalty payment from a franchisee to the parent org when the monthly period closes.

Pre-conditions	Franchise DAO active. Royalty smart contract deployed. Franchisee Finance period closes.
Post-conditions	Royalty paid automatically. Both orgs' Finance modules updated. DLT block committed. No manual intervention required.

Main Flow

1	Franchisee OrgOS Finance module emits finance.period.closed event at month end.
2	DAO inter-org event bus routes event to smart contract engine.
3	Smart contract evaluates royalty term: royalty_rate × period gross revenue.
4	Smart contract calculates payment amount from franchisee Financial data (authorised read via DAO data sharing ODD).
5	Smart contract initiates treasury transfer: debits franchisee treasury, credits parent treasury.
6	Both OrgOS instances update their Finance modules: franchisee records royalty expense JE; parent records royalty income JE.
7	ROYALTY_PAYMENT transaction committed to DLT with amount, from/to orgs, period reference, and contract ID.
8	Both orgs notified. DAO dashboard treasury balances update.

Alternate / Exception Flows

Insufficient franchisee treasury	Smart contract engine flags payment failure. DAO governance notified. Cure period contract term activated.
Period dispute	Franchisee disputes revenue figure. DAO dispute resolution process initiated. Payment held until resolved.

Group J — Marketing & Work Management

UC-47 Generate AI Campaign Content [Marketing]

Actor	Marketing Manager / Owner
Goal	Use the AI Campaign Generator to produce ready-to-publish social and email content for a target persona.
Pre-conditions	Marketing OrgApp active. At least one micro-persona defined.
Post-conditions	Content approved, scheduled, and published. Engagement tracking active.

Main Flow

1	User opens Marketing → AI Campaign Generator.
2	Selects target micro-persona, campaign goal, brand tone, and channels.
3	Clicks "Generate Campaign Content".
4	AI generates: post copy for each channel, hashtags, email subject line, email body, and campaign tagline.
5	User reviews generated content. Can regenerate individual pieces or all.
6	User edits content inline. Approves final versions.
7	Clicks "Schedule". Content pushed to Content Calendar with selected dates.
8	System publishes via connected social/email drivers at scheduled time. Engagement metrics collected.

Alternate / Exception Flows

Brand voice deviation	User flags a generated piece as off-brand. Feedback stored to improve future generation for this OrgOS.

UC-48 Create and Complete a Work Task [Work / ODD]

Actor	Any OrgExec (Human or AI)
Goal	Create a task on the work board, assign it, progress it through to completion.

Pre-conditions	Work Management OrgApp active. User has Work Write permission.
Post-conditions	Task completed. Completion recorded. Work analytics updated. Any cross-module triggers fired (e.g., task completion triggers invoice milestone).

Main Flow

1	User opens Work → Sprint Board → "+ New Task".
2	Enters task title, description, due date, priority.
3	Assigns to an OrgExec. System checks assignee's ODD: confirms they have capability to perform this task type.
4	Task appears on assignee's board in Backlog. Assignee notified.
5	Assignee moves task to In Progress when work starts.
6	Assignee completes work. Moves task to Review (if review required per workflow config) or Done.
7	If review: reviewer approves or returns with feedback.
8	On Done: completion timestamp recorded. Work analytics updated. Any linked KPIs refreshed.

Alternate / Exception Flows

ODD capability mismatch	If assigned OrgExec's ODD does not include this task type as a capability, system warns user. Can override with justification.
Task overdue	At due date: task card flagged amber. Manager notified. Overdue duration tracked for Work KPI.
AI agent task	If assigned to AI OrgExec, agent picks up task via task queue API. Progress updates automatically.

Part III: User Workflows

User workflows describe complete end-to-end business processes as they flow through the UME OrgOS. Each workflow shows the full sequence of steps across actors, phases, and modules — including system automations, cross-module event cascades, and AI OrgExec actions. Workflows are grouped by the business process domain they represent.

Group	Workflows	Domain
1 — Business Setup & Onboarding	WF-01 to WF-04	OrgOS setup, entity, team onboarding
2 — Order-to-Cash	WF-05 to WF-09	Sales, fulfilment, invoicing, payment
3 — Procure-to-Pay	WF-10 to WF-13	Procurement, receiving, AP, payment
4 — Hire-to-Retire	WF-14 to WF-18	HR lifecycle: hire, onboard, leave, review, offboard
5 — Compliance & Risk	WF-19 to WF-22	Compliance filings, risk response, audits
6 — Finance Close	WF-23 to WF-25	Period close, payroll, consolidated reporting
7 — AI Agent Operations	WF-26 to WF-28	AI OrgExec deployment, escalation, monitoring
8 — DAO & Franchise Operations	WF-29 to WF-32	DAO formation, royalties, OrgDNA, governance

Workflow Legend

User Action — A step performed directly by a human user or OrgExec.
System — An automated action performed by the OrgKernel or an OrgApp.
Decision — A conditional branch point where the flow diverges.
Automation — An action triggered by an event, timer, or AI OrgExec.
External — An action performed outside UME (e.g., filing with an authority, customer paying).

Group 1 — Business Setup & Onboarding

WF-01 Launch a New OrgOS from Template

Trigger	New user creates a UME account and wants a running OrgOS within one session.
Actors	New User (Owner/Founder)
Modules	Auth · OrgOS Provisioner · Legal · Finance · Dashboard

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	New User	Visits ume.io, enters name, org name, email, password. Accepts Terms.
2	**System**	Auth	Sends 6-digit OTP email. Awaits verification.
3	**User Action**	New User	Enters OTP. Account verified.
4	**System**	OrgOS Provisioner	Presents Template Gallery (12+ industry templates). User selects closest match.
5	**User Action**	New User	Reviews pre-selected modules in Module Configurator. Toggles adjustments. Pricing updates live.
6	**User Action**	New User	Enters legal entity type, registered address, jurisdiction, and tax ID.
7	**System**	Legal / Chombo	Fetches compliance framework for jurisdiction. Pre-generates compliance calendar.
8	**User Action**	New User	Optionally invites first team members with roles.
9	**User Action**	New User	Clicks "Launch OrgOS".
10	**System**	OrgOS Provisioner	Provisions OrgKernel, all selected OrgApps, DB schema namespaces, COA, compliance calendar, default ODDs.
11	**System**	OrgOS Provisioner	Displays step-by-step boot progress log in real time. All steps tick green.
12	**System**	OrgDNA Manager	Commits OrgDNA v1.0.0 for this new OrgOS instance.
13	**System**	Dashboard	Renders Executive Dashboard. Compliance calendar widget shows first upcoming obligation.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`orgos.provisioned`	SDO Engine (initialises digital twin), Analytics (creates baseline KPIs)
`legal.compliance_calendar.generated`	Dashboard (compliance widget), Risk module (populates initial KRI set)
`orgdna.version.created`	Audit log, OrgDNA Manager
`auth.user.account_created`	Audit log, Welcome email via CommDriver

Success Criteria

- ✓ OrgOS fully provisioned and all selected OrgApps in Running state.
- ✓ Compliance calendar populated with all jurisdiction-specific obligations.
- ✓ Chart of accounts generated from industry template.
- ✓ OrgDNA v1.0.0 committed. SDO digital twin initialised.
- ✓ User is on Executive Dashboard within 90 seconds of clicking "Launch OrgOS".

WF-02 Onboard a New Employee End-to-End

Trigger	HR Manager initiates new hire process for an approved position.
Actors	HR Manager → New Employee → IT Admin (automated)
Modules	HR · Legal · Finance · IT · Portal · ODD

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	HR Manager	Opens HR → People → New Hire wizard. Enters personal info, role, department, OrgExec type: Human.
2	**User Action**	HR Manager	Selects ODD template for role (e.g., FTE-ODD-v2). Reviews auto-populated capabilities and constraints.
3	**User Action**	HR Manager	Uploads right-to-work documents. Enters salary, pay schedule, benefits selections.
4	**User Action**	HR Manager	Reviews summary screen. Confirms. Clicks "Create Employee".
5	**System**	HR Module	Creates employee record. Registers OrgExec in ODDBus with generated ODD. Emits hr.employee.created.
6	**Automation**	Legal Module	Receives event. Generates employment contract from template. Sends to employee for e-signature.
7	**Automation**	Finance Module	Receives event. Creates payroll account. Adds to next payroll run.
8	**Automation**	IT Module	Receives event. Generates provisioning request for email, Slack, system access per ODD data access grants.
9	**Automation**	Portal Module	Receives event. Creates employee portal account. Sends welcome email with login link.
10	**External**	New Employee	Signs employment contract via e-sign link.
11	**System**	Legal Module	Receives signature confirmation. Stores executed contract in document vault. Emits legal.contract.executed.
12	**System**	HR Module	Timeline entry: "Employment contract signed". Employee status → Active.
13	**Automation**	SDO Engine	Adds new OrgExec node to digital twin topology.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`hr.employee.created`	Legal (contract), Finance (payroll), IT (provisioning), Portal (welcome email), SDO (topology update)
`legal.contract.executed`	HR (timeline entry), Document Vault (archive), Audit log
`odd.contract.registered`	ODDBus (enforce from first login), Audit log
`it.provisioning.requested`	IT Module (queues provisioning tasks), connected system drivers

Success Criteria

- ✓ Employee record active with ODD registered and enforced.
- ✓ Employment contract executed and stored in document vault.
- ✓ Payroll account created and included in next payroll cycle.
- ✓ Portal access granted. Employee can log in with assigned ODD-scoped permissions.
- ✓ All five cross-module cascade steps completed within 30 seconds of "Create Employee".

WF-03 Register an Entity and Generate Compliance Calendar

Trigger	Owner setting up a new legal entity within existing OrgOS.
Actors	Owner / Legal Officer
Modules	Legal / Chombo · Dashboard · Risk

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Owner	Opens Legal & Compliance → Entities → New Entity.
2	**User Action**	Owner	Enters entity name, type (LLC/Ltd/etc.), registration number, registered address, jurisdiction.
3	**System**	Legal / Chombo	Queries compliance framework database for entity type + jurisdiction combination.
4	**System**	Legal / Chombo	Generates draft compliance calendar: all statutory filings, annual returns, tax deadlines, licence renewals.

5	**User Action**	Owner	Reviews generated obligations. Suppresses any not applicable. Adds any custom obligations.
6	**User Action**	Owner	Links entity to Finance (assigns COA entity tag) and HR (assigns headcount pool).
7	**User Action**	Owner	Saves entity. Calendar activated.
8	**System**	Dashboard	Compliance calendar widget updates. Next obligation for this entity shown with countdown.
9	**System**	Risk Module	Receives event. Adds jurisdiction-specific compliance risk entries to risk register template.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`legal.entity.registered`	Finance (entity COA tag), HR (entity headcount pool), Dashboard (compliance widget)
`legal.compliance_calendar.generated`	Risk module (compliance risk entries), Dashboard, Notification service (upcoming deadline alerts)

Success Criteria

- ✓ Entity registered with all fields validated.
- ✓ Compliance calendar populated — all filing obligations for entity type and jurisdiction present.
- ✓ Entity linked to Finance and HR modules.
- ✓ Dashboard compliance widget reflects new entity obligations.

WF-04 OrgDNA Export and Franchise Clone

Trigger	Franchise owner wants to spin up a new franchisee OrgOS from their proven config.
Actors	Franchise Owner / Admin
Modules	OrgDNA Manager · OrgOS Provisioner · DAO

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Franchise Owner	Opens OrgSystem → OrgDNA Manager → Export OrgDNA.
2	**System**	OrgDNA Manager	Serialises full OrgOS config: kernel config, all OrgApp settings, ODD templates, COA, compliance packs, RBAC.
3	**System**	OrgDNA Manager	Runs validation. Confirms all dependencies satisfied. No transactional data included.
4	**System**	OrgDNA Manager	Produces versioned OrgDNA package. Stores in OrgDNA vault. Ready for export or clone.
5	**User Action**	Franchise Owner	Clicks "Clone to New OrgOS". Enters franchisee org name, registered address, jurisdiction.
6	**User Action**	Franchise Owner	Adjusts jurisdiction-specific compliance packs for franchisee location. Reviews COA if any local adjustments needed.
7	**User Action**	Franchise Owner	Confirms clone. Clicks "Provision New OrgOS".
8	**System**	OrgOS Provisioner	Provisions new OrgOS instance from cloned OrgDNA: new org_id, isolated DB namespace, all OrgApps booted.
9	**System**	OrgDNA Manager	Commits OrgDNA v1.0.0 on new instance. Records parent OrgDNA lineage (parent_dna_id set).
10	**System**	Notification	Sends franchisee access credentials email. New OrgOS live and accessible.
11	**Automation**	DAO Module	If franchise DAO exists: automatically sends DAO join invitation to new OrgOS. Franchisee admin receives invite notification.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`orgdna.version.created`	OrgDNA vault (archive), Audit log

`orgos.provisioned`	New OrgOS SDO Engine, New OrgOS Analytics, Notification service
`dao.join.invitation_sent`	Franchisee OrgOS DAO module (shows pending invitation)

Success Criteria

- ✓ New OrgOS live within 90 seconds of confirming provisioning.
- ✓ All franchise-standard OrgApps running with correct configuration cloned from parent.
- ✓ Franchisee has full independent OrgOS — operationally sovereign, brand-aligned.
- ✓ DAO join invitation sent if franchise DAO exists.
- ✓ OrgDNA parent-child lineage recorded for brand standards tracking.

Group 2 — Order-to-Cash

WF-05 Dry Cleaning Order — Full Cycle (Bob & Alice)

Trigger	Customer drops off garments for dry cleaning at the counter.
Actors	Front Desk Staff · Cleaning OrgCPU (washer/dryer) · Customer (external) · Finance
Modules	CRM / Sales · Operations · Finance · Communications

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Front Desk Staff	Opens CRM → Order Board → New Order. Searches for customer or creates new contact.
2	**User Action**	Front Desk Staff	Enters items (shirt × 3, suit × 1), special instructions, and collection date. Price auto-calculated.
3	**System**	CRM Module	Creates order. Card placed in "Received" column of kanban. Order ticket printed if printer connected.
4	**User Action**	Staff	Tags garments and moves to cleaning queue. Moves kanban card to "Cleaning".
5	**Automation**	Scheduling AI OrgExec	Scheduling agent reads order queue. Assigns optimal machine (OrgCPU) slot based on current load and ETA.
6	**System**	OrgCPU (Washer)	Machine processes garments. Reports cycle status via IoT ODD to Operations dashboard.
7	**User Action**	Staff	Garments cleaned and pressed. Moves kanban card to "Ready".
8	**Automation**	CRM / Comms	Status change to "Ready" triggers automated SMS to customer: "Your order is ready for collection."
9	**External**	Customer	Customer arrives and collects garments.
10	**User Action**	Staff	Confirms collection. Moves order to "Collected". System auto-generates invoice.
11	**Automation**	Finance Module	Receives order.collected event. Posts revenue journal entry. Updates AR or records payment if paid in-store.
12	**System**	Analytics	Order KPI updated: average turnaround time, revenue per order, machine utilisation.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`sales.order.created`	Operations (schedule slot), Analytics (order count KPI)
`sales.order.status_changed → Ready`	CommDriver (sends customer SMS notification)
`sales.order.collected`	Finance (revenue JE), CRM (customer history update), Analytics (revenue KPI)
`orgcpu.cycle.completed`	Operations dashboard (machine utilisation), SDO (OrgCPU node update)

Success Criteria

- ✓ Order fulfilled and collected. Customer notified automatically at each status change.
- ✓ Revenue posted to Finance automatically on collection.
- ✓ Machine utilisation recorded. Scheduling agent improves future assignments from this data.
- ✓ Customer history updated for future loyalty and marketing use.

WF-06 B2B Sales Deal — Lead to Invoice

Trigger	Sales rep qualifies a new inbound lead for a B2B services engagement.
Actors	Sales Rep · Sales Manager · Finance
Modules	CRM / Sales · Finance · Legal · Work

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Sales Rep	Creates new Deal in CRM pipeline. Links to company contact. Sets estimated value and close date. Stage: Qualified.
2	**User Action**	Sales Rep	Logs discovery call activity. Updates deal notes. Moves deal to "Proposal" stage.
3	**User Action**	Sales Rep	Creates proposal using Legal document template. Sends via e-sign integration.
4	**External**	Customer	Customer reviews proposal and signs.
5	**Automation**	Legal Module	Detects e-sign completion. Emits legal.contract.executed. Stores signed proposal in document vault.
6	**System**	CRM Module	Receives contract event. Moves deal to "Won". Creates project/work order linked to deal.
7	**Automation**	Finance Module	Receives deal.won event. Creates billing schedule based on contract terms (milestone or recurring).
8	**User Action**	Sales Rep/Manager	Reviews auto-created billing milestones. Confirms first invoice details.
9	**System**	Finance Module	Generates and sends first invoice. Updates pipeline dashboard (revenue recognised).
10	**System**	Work Module	Creates project tasks from contract scope. Assigns to delivery team OrgExecs.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`crm.deal.won`	Finance (billing schedule), Work (project tasks), Analytics (win rate KPI)
`legal.contract.executed`	CRM (deal status update), Finance (contract billing reference), Document Vault
`finance.invoice.sent`	CRM (deal revenue update), Analytics (revenue pipeline)

Success Criteria

- ✓ Deal won and contract executed. Stored in document vault.
- ✓ Billing schedule created matching contract terms.
- ✓ First invoice sent within the session.
- ✓ Project tasks created and assigned to delivery team.
- ✓ Sales pipeline and forecast updated automatically.

WF-07 E-commerce Order — OrgCPU (POS) to Fulfilment

Trigger	Customer makes a purchase via POS terminal (OrgCPU) in physical retail store.
Actors	Customer (external) · POS Terminal (OrgCPU) · Inventory · Finance
Modules	CRM · Operations / Inventory · Finance · Communications

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**External**	Customer	Customer presents items at POS terminal. Staff processes sale.
2	**System**	POS OrgCPU	POS records sale. IOT-ODD governs data reporting: emits sale event with items, amounts, payment method.
3	**Automation**	Operations Module	Receives sale event. Decrements inventory quantities for sold items. Checks reorder points.
4	**Decision**	Operations Module	Are any sold items now at or below reorder point?
5	**Automation**	Operations Module	IF YES: creates PO draft for replenishment. Alerts Purchasing OrgExec.
6	**Automation**	Finance Module	Receives sale event. Posts revenue journal entry. Updates daily cash position.
7	**Automation**	CRM Module	If customer identified (loyalty card): updates purchase history. Evaluates re-engagement campaign eligibility.

8	System	Analytics	Updates real-time sales KPIs: revenue today, units sold, top SKUs, margin by product.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`orgcpu.sale.recorded`	Operations (inventory decrement), Finance (revenue JE), CRM (customer history), Analytics (KPI update)
`operations.inventory.low_stock`	Operations (auto-draft PO), Dashboard (low stock alert), Risk (KRI update if critical item)

Success Criteria

- ✓ Sale recorded and revenue posted in real time via OrgCPU ODD integration.
- ✓ Inventory immediately updated. Reorder triggered if threshold breached.
- ✓ Customer history updated for marketing segmentation.
- ✓ Finance GL reflects sale within one event propagation cycle.

WF-08 Invoice Payment — Customer Pays via Stripe

Trigger	Customer clicks payment link in invoice email and completes payment.
Actors	Customer (external) · Stripe (external) · Finance Module (automation)
Modules	Finance · CRM · Dashboard

Workflow Steps

Step | Phase | Actor | Action / System Response

1	External	Customer	Clicks payment link in invoice email. Stripe checkout page loads with invoice amount.
2	External	Customer	Enters card details. Submits payment.
3	External	Stripe	Processes payment. Sends webhook to UME Finance driver.
4	System	Finance Driver	Receives Stripe webhook. Validates amount and invoice reference.
5	Automation	Finance Module	Marks invoice as Paid. Posts cash receipt journal entry (Debit: Bank, Credit: AR).
6	Automation	Finance Module	Creates bank reconciliation item. Updates AR ageing report. Recalculates cash flow forecast.
7	Automation	CRM Module	Receives payment event. Updates order/deal record as Paid. Clears any overdue AR flag on customer record.
8	Automation	Dashboard	Cash position widget updates. AR outstanding balance reduces. Revenue chart updates.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.invoice.paid`	CRM (order status paid), AR ageing (balance update), Cash flow forecast (update), Analytics (cash KPI)
`finance.journal_entry.posted`	Audit log, SDO (financial state update)

Success Criteria

- ✓ Invoice marked Paid within seconds of Stripe webhook receipt.
- ✓ Cash receipt JE posted. AR balance updated. Bank reconciliation item created.
- ✓ Customer record and order history updated automatically.
- ✓ No manual intervention required from payment initiation to GL posting.

WF-09 Accounts Receivable Ageing and Collections

Trigger	Finance Manager reviews overdue invoices at end of month and initiates collection actions.
Actors	Finance Manager · Customer (external)

Modules	Finance · CRM · Communications

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Finance Manager	Opens Finance → AR Ageing. Reviews outstanding invoices grouped by age bucket.
2	**User Action**	Finance Manager	Filters to 30+ days overdue. Selects priority invoices by amount.
3	**User Action**	Finance Manager	For each priority invoice: clicks "Send Reminder". System sends payment reminder email.
4	**External**	Customer	Customer receives reminder. Some pay via link. Others contact to discuss.
5	**Decision**	Finance Manager	Is invoice 60+ days overdue without payment commitment?
6	**User Action**	Finance Manager	IF YES: marks invoice "In Collections". Logs collection action. Adds note to customer CRM record.
7	**Automation**	Finance Module	Creates bad debt provision entry if 90+ days (if auto-provision configured).
8	**Automation**	Risk Module	Receives collection escalation. Updates AR concentration KRI if single customer represents large overdue balance.
9	**User Action**	Finance Manager	Runs monthly AR report. Presents to owner with ageing summary and recovery forecast.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.invoice.overdue`	Risk module (KRI update), Dashboard (AR ageing alert), CRM (customer credit flag)
`finance.ar.collections_escalated`	Risk module (KRI), CRM (customer record flag), Audit log

Success Criteria

- ✓ All overdue invoices actioned with reminders sent.
- ✓ Collections cases escalated with documentation.
- ✓ Risk module KRI updated to reflect AR position.
- ✓ Monthly AR report available for management review.

Group 3 — Procure-to-Pay

WF-10 Purchase Order — Raise to Receive

Trigger	Inventory falls below reorder point. Purchasing manager raises a PO and receives goods.
Actors	Operations Manager · Supplier (external) · Finance
Modules	Operations · Finance · Inventory · Communications

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**Automation**	Operations Module	Inventory check detects stock below reorder point. Creates PO draft with preferred vendor and recommended quantity.
2	**System**	Dashboard	Low-stock alert appears in Operations widget. Finance widget shows projected AP liability.
3	**User Action**	Operations Manager	Reviews PO draft. Adjusts quantity if needed. Selects delivery date.
4	**User Action**	Operations Manager	Submits PO. If above ODD approval threshold: routes to Finance Manager for approval.
5	**Automation**	Communications	PO confirmation sent to supplier by email (or EDI if connected).
6	**External**	Supplier	Supplier confirms PO. Dispatches goods.
7	**User Action**	Operations Manager	Goods arrive. Opens PO → Receive Goods. Enters quantities received. Flags any discrepancies.
8	**System**	Operations Module	Updates inventory quantities. Marks PO as Received. Emits operations.po.goods_received.
9	**Automation**	Finance Module	Receives goods_received event. Creates AP invoice record. Awaits supplier invoice for three-way match.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`operations.inventory.low_stock`	Operations (draft PO), Dashboard (alert), Risk (KRI if critical item)
`operations.po.submitted`	CommDriver (supplier email), Audit log, Finance (AP accrual)
`operations.po.goods_received`	Inventory (quantity update), Finance (AP invoice pending), Analytics (inventory KPI)

Success Criteria

- ✓ Stock replenished. Inventory quantities updated in real time.
- ✓ PO fully documented from draft to receipt.
- ✓ AP record created and matched to PO receipt. Ready for supplier invoice.
- ✓ No manual inventory counting required — OrgCPU sensors update automatically where connected.

WF-11 Supplier Invoice — Three-Way Match and Payment

Trigger	Supplier sends invoice for delivered goods. Finance processes payment.
Actors	Finance / AP · Supplier (external)
Modules	Finance · Operations

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**External**	Supplier	Supplier emails invoice (PDF or EDI). Finance receives.
2	**User Action**	Finance AP	Opens Finance → AP → New Supplier Invoice. Enters or imports invoice details.
3	**System**	Finance Module	Runs three-way match: PO quantities ↔ goods receipt quantities ↔ invoice quantities and amounts.
4	**Decision**	Finance Module	Does three-way match pass within tolerance?

5	Automation	Finance Module	IF PASS: invoice auto-approved. Scheduled for payment on due date.
6	User Action	Finance Manager	IF FAIL: Finance Manager reviews discrepancy report. Contacts supplier to resolve. Approves manually once resolved.
7	System	Finance Module	On due date: payment processed via Banking driver. Journal entry posted (Debit: AP, Credit: Bank).
8	System	Finance Module	Invoice marked Paid. Bank reconciliation item created. AP ageing updated.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.ap.invoice_approved`	Finance (payment schedule), Audit log
`finance.ap.payment_processed`	Finance (GL post), Bank reconciliation (match item), Analytics (cash flow update)

Success Criteria

- ✓ Supplier paid on time. No manual matching errors.
- ✓ Three-way match documented in audit trail.
- ✓ AP ageing and cash flow forecast updated automatically.

WF-12 Expense Claim — Staff Submits and Finance Approves

Trigger	Employee incurs a work expense and submits for reimbursement.
Actors	Employee (OrgExec) · Finance Manager
Modules	Finance · HR · Portal

Workflow Steps

Step | Phase | Actor | Action / System Response

1	User Action	Employee	Opens Employee Portal → Expenses → New Claim. Enters amount, category, date, description.
2	User Action	Employee	Uploads photo of receipt. Submits claim.
3	System	Finance Module	Validates claim against employee ODD (expense category permitted, amount within per-claim limit).
4	Decision	Finance Module	Does amount exceed the employee ODD approval threshold?
5	Automation	Finance Module	IF BELOW: auto-approves. Posts accrual entry. Schedules for next payroll run.
6	Automation	Finance Module	IF ABOVE: routes to Finance Manager for approval. Notification sent.
7	User Action	Finance Manager	Reviews claim and receipt. Approves or declines with comment.
8	Automation	Finance Module	On approval: posts expense JE. Includes in next payroll payment. Employee notified.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.expense.submitted`	Finance (ODD validation, approval routing), Audit log
`finance.expense.approved`	Payroll module (include in next run), Employee portal (status update), Audit log

Success Criteria

- ✓ Expense reimbursed in next payroll cycle.
- ✓ Receipt stored. Expense category and ODD compliance documented.
- ✓ Finance GL updated. No lost receipts or manual spreadsheets.

WF-13 Contract Renewal — Legal to Finance

Trigger	A vendor contract is approaching expiry. Legal flags for renewal. Finance is notified of rate change.

Actors	Legal Officer · Owner / Finance Manager
Modules	Legal · Finance · Communications

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**Automation**	Legal Module	Contract expiry date triggers 60-day renewal reminder. Compliance calendar entry highlighted.
2	**System**	Notification Service	Sends renewal alert to Legal Officer and Owner.
3	**User Action**	Legal Officer	Reviews contract. Decides to renew. Contacts vendor to negotiate terms.
4	**External**	Vendor	Vendor provides updated contract with new rate.
5	**User Action**	Legal Officer	Updates contract record in Legal vault: new rates, new expiry date. Generates renewal doc from template.
6	**User Action**	Legal Officer	Sends for e-signature. Both parties sign.
7	**Automation**	Legal Module	Detects e-sign completion. Archives renewed contract. Emits legal.contract.renewed.
8	**Automation**	Finance Module	Receives renewal event. Updates recurring AP entry with new rate. Alerts Finance Manager of rate change.
9	**System**	Legal Module	Sets new 60-day renewal reminder at new expiry date.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`legal.contract.renewal_due`	Notification service (alert to Legal Officer and Owner), Dashboard (compliance item)
`legal.contract.renewed`	Finance (AP rate update), Document Vault (archive new contract), Audit log

Success Criteria

- ✓ Contract renewed before expiry. No service interruption.
- ✓ New rates updated in Finance AP automatically.
- ✓ Next renewal reminder set. No risk of missing expiry.

Group 4 — Hire-to-Retire

WF-14 Annual Leave Request and Approval

Trigger	Employee requests annual leave through the portal.
Actors	Employee (OrgExec) · Line Manager
Modules	HR · Portal · Operations · Payroll

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Employee	Opens Employee Portal → Leave → New Leave Request. Selects type: Annual Leave.
2	**User Action**	Employee	Selects date range. System shows current balance (18 days remaining) and balance after approval (14 days).
3	**System**	HR Module	Validates request: sufficient balance, no blackout dates, no scheduling conflict with other approved leave.
4	**User Action**	Employee	Adds optional message to manager. Submits.
5	**Automation**	HR Module	Notification sent to Line Manager: "Alice has requested leave [dates]. Action required."
6	**User Action**	Line Manager	Opens HR → Leave Inbox. Reviews request alongside team leave calendar and order queue.
7	**Decision**	Line Manager	Can the team absorb the absence for these dates?
8	**User Action**	Line Manager	IF YES: Approves. IF NO: Declines with suggested alternative dates.
9	**Automation**	HR Module	On approval: deducts from leave balance, updates team calendar, notifies employee by email.
10	**Automation**	Operations Module	Receives leave.approved event. Marks staff member unavailable in scheduling calendar for those dates.
11	**Automation**	Payroll Module	Receives leave.approved event. Records annual leave days for payroll period calculation.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`hr.leave.requested`	Notification service (manager alert), Audit log
`hr.leave.approved`	Employee portal (balance update, confirmation email), Operations (schedule update), Payroll (period record)
`hr.leave.declined`	Employee portal (decline notification with reason)

Success Criteria

- ✓ Leave approved or declined promptly. Employee notified.
- ✓ Leave balance updated. Schedule updated. Payroll informed.
- ✓ Manager had full context (team calendar) to make informed decision.

WF-15 Payroll Run — End-to-End

Trigger	Monthly payroll processing for all active employees.
Actors	HR / Payroll Admin · Finance Manager
Modules	HR · Finance · Portal · Legal

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**System**	HR / Payroll	At configured payroll date: system generates payroll draft for all active employees with contracted salaries.
2	**Automation**	HR Module	Imports approved timesheets (for hourly staff), approved leave (for AL days), approved expenses for inclusion.

3	**User Action**	Payroll Admin	Reviews payroll draft. Checks each employee: gross pay, deductions (tax, NI/FICA, pension), net pay.
4	**User Action**	Payroll Admin	Enters any manual adjustments: bonuses, overtime, commission payments.
5	**Decision**	Payroll Admin	Does total payroll exceed dual-sign threshold (per ODD)?
6	**User Action**	Finance Manager	IF YES: Finance Manager reviews and co-approves payroll run.
7	**User Action**	Payroll Admin	Confirms payroll run. Clicks "Process Payroll".
8	**System**	Finance Module	Posts payroll journal entry: Gross Wages Dr, Tax Liability Cr, Net Wages Payable Cr, Pension Contribution Cr.
9	**Automation**	HR / Portal	Generates payslips for all employees. Payslips published to employee portal. Email notification sent.
10	**Automation**	Legal / Chombo	Tax liability and payroll tax filing due date added to compliance calendar.
11	**System**	Finance Module	Net wages payment batch queued for banking driver. Processed on configured payment date.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`hr.payroll.run_confirmed`	Finance (GL posting), Portal (payslips), Legal (tax liability calendar entry), Audit log
`finance.journal_entry.posted`	Audit log, SDO (financial state update)
`hr.payroll.tax_due`	Compliance calendar, Notification service (upcoming tax payment reminder)

Success Criteria

- ✓ All employees paid correctly and on time.
- ✓ Payslips available in portal within minutes of processing.
- ✓ Payroll journal entry posted. Tax liability in compliance calendar.
- ✓ Full audit trail: who ran payroll, who approved (if dual-sign), every line item.

WF-16 Employee Offboarding

Trigger	An employee resigns. HR initiates offboarding process.
Actors	HR Manager · IT Admin (automated) · Finance
Modules	HR · Legal · Finance · IT · Portal

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	HR Manager	Opens employee record → Offboard Employee. Enters last day, reason, notice period status.
2	**System**	HR Module	Generates offboarding checklist: exit interview, equipment return, final payslip, access revocation, legal docs.
3	**User Action**	HR Manager	Schedules exit interview. Records outcomes in HR system.
4	**Automation**	Legal Module	Generates separation agreement (if applicable) from template. Sends for e-signature.
5	**User Action**	HR Manager	Works through checklist: equipment return confirmed, exit interview completed, final payslip reviewed.
6	**Automation**	Finance Module	Calculates final pay: outstanding salary, accrued holiday pay, any statutory payments. Posts to next payroll.
7	**System**	HR Module	On last day: emits hr.employee.offboarded event.
8	**Automation**	IT Module	Receives offboard event. Triggers deprovisioning of all system access via device drivers (email, Slack, other SSO apps).
9	**Automation**	ODDBus	Suspends employee ODD. Employee can no longer perform any OrgOS actions.
10	**System**	HR Module	Employee status → Inactive. Record archived with retention policy applied.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`hr.employee.offboarded`	IT (deprovisioning), ODDBus (ODD suspension), Finance (final pay), Portal (access revocation), Audit log
`odd.contract.suspended`	ODDBus (immediate enforcement), Audit log
`it.deprovisioning.completed`	HR (checklist item tick), Audit log

Success Criteria

- ✓ Employee access revoked on last day. ODD suspended. No orphaned accounts.
- ✓ Final pay calculated and included in next payroll.
- ✓ Separation documents signed and archived.
- ✓ Full offboarding audit trail maintained.

WF-17 Performance Review Cycle

Trigger	Annual performance review cycle initiated by HR for all staff.
Actors	HR Manager · Line Manager · Employee
Modules	HR · Work · Portal

Workflow Steps

Step | Phase | Actor | Action / System Response

Step	Phase	Actor	Action / System Response
1	**User Action**	HR Manager	Opens HR → Performance → Start Review Cycle. Selects cycle type (Annual), sets dates, selects participants.
2	**System**	HR Module	Creates review records for all included employees. Assigns to their line managers.
3	**Automation**	Portal	Employees receive portal notification to complete self-assessment by [date].
4	**User Action**	Employee	Completes self-assessment: rates goals, competencies, achievements. Submits.
5	**Automation**	HR Module	Notifies line manager that self-assessment is ready for review.
6	**User Action**	Line Manager	Completes manager assessment. Optionally requests 360 feedback from peers.
7	**User Action**	Line Manager	Holds review meeting. Records meeting notes and agreed development actions.
8	**User Action**	Line Manager	Submits final review. Review shared with employee on portal.
9	**User Action**	Employee	Acknowledges review via portal.
10	**Automation**	HR Module	On all reviews completed: emits hr.review_cycle.completed. Analytics updated with aggregate ratings.
11	**Decision**	HR Manager	Does review outcome trigger a compensation review?
12	**User Action**	HR Manager	IF YES: initiates compensation adjustment workflow for affected employees.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`hr.review_cycle.started`	Portal (notifications to employees and managers), Audit log
`hr.review.completed`	HR Analytics (aggregate ratings update), Portal (employee acknowledgement prompt)
`hr.review_cycle.completed`	HR Analytics (cycle summary), Compensation workflow (if triggered)

Success Criteria

- ✓ All employees reviewed within the cycle window.
- ✓ Review records stored with full history.
- ✓ Development actions assigned as Work tasks.
- ✓ HR analytics updated with ratings data for compensation and succession planning.

WF-18 Sick Leave and Return-to-Work

Trigger	Employee calls in sick. Manager records absence. Employee returns and follows return process.
Actors	Employee · Line Manager · HR
Modules	HR · Portal · Operations · Payroll

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Line Manager	Records absence: HR → Absence → Log Absence. Selects employee, date, type: Sick.
2	**Automation**	Operations Module	Receives absence event. Marks staff member unavailable in scheduling. Notifies Scheduling AI OrgExec to re-route assignments.
3	**Automation**	Scheduling AI	Re-assigns affected tasks/orders to available OrgExecs within their ODD capability constraints.
4	**Automation**	HR Module	Tracks consecutive sick days. After threshold (default: 3 days): flags for manager follow-up. After 7 days: flags for HR review.
5	**External**	Employee	Employee recovers. Returns to work.
6	**User Action**	Line Manager	Records return. Opens return-to-work interview template in HR. Completes and stores interview record.
7	**System**	HR Module	Absence record closed. Duration calculated. Sick leave balance updated. Payroll notified for statutory sick pay calculation if applicable.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`hr.absence.recorded`	Operations (schedule update), Scheduling AI (re-assignment), Payroll (SSP calculation if applicable)
`hr.absence.returned`	Operations (reinstate in schedule), HR Analytics (absence rate KPI), Payroll (end SSP if applicable)

Success Criteria

- ✓ Absence managed with minimal operational disruption.
- ✓ Scheduling AI re-routed work automatically within ODD constraints.
- ✓ Return-to-work interview documented.
- ✓ Payroll notified for correct statutory pay treatment.

Group 5 — Compliance & Risk

WF-19 Tax Filing — Quarterly VAT / Sales Tax

Trigger	Quarterly VAT/sales tax filing due. System reminds. Owner completes and records.
Actors	Owner / Finance Manager
Modules	Legal / Chombo · Finance · Dashboard

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**Automation**	Compliance Calendar	14 days before due: notification sent to Owner and Finance Manager. Compliance calendar item turns amber.
2	**User Action**	Finance Manager	Opens Legal → Filing Calendar. Clicks on VAT/sales tax filing entry.
3	**System**	Legal / Chombo	Shows filing detail: description, due date, auto-calculated tax amount from Finance transactions in the period.
4	**User Action**	Finance Manager	Reviews calculated amount. Reconciles against Finance reports if needed.
5	**External**	Finance Manager	Logs into tax authority portal. Submits return with the calculated amount. Obtains confirmation reference.
6	**User Action**	Finance Manager	Returns to UME. Clicks "Mark as Filed". Enters filing date and reference. Uploads confirmation document.
7	**System**	Legal / Chombo	Marks obligation Complete. Stores evidence in document vault. Creates next quarterly occurrence.
8	**Automation**	Dashboard	Compliance score updated. Item removed from overdue/due-soon list. Next occurrence appears in calendar.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`legal.filing.completed`	Compliance calendar (next occurrence), Document vault (evidence), Dashboard (compliance score update), Audit log

Success Criteria

- ✓ Filing submitted on time with evidence stored.
- ✓ Next quarter occurrence auto-scheduled.
- ✓ Compliance score reflects successful filing.
- ✓ Audit trail: who filed, when, what reference — immutable.

WF-20 Risk KRI Breach — Detection to Response

Trigger	A key risk indicator breaches its threshold. System detects and routes to responsible executive.
Actors	OrgKernel (automatic) · Risk Manager · Executive
Modules	Risk / GRC · Dashboard · Work · Notification

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**Automation**	Risk Module	KRI monitor evaluates cash runway KRI: current value = 18 days. Threshold = 30 days. BREACH detected.
2	**System**	Risk Module	Creates KRI breach record. Emits risk.kri.breach event. Updates SDO risk layer.
3	**Automation**	Dashboard	Executive Dashboard shows red KRI breach card: "Cash Runway — 18 days (threshold: 30 days)". Sparkline shows declining trend.
4	**Automation**	Notification	Pushes alert to Owner and Finance Manager: "KRI Breach: Cash Runway."
5	**User Action**	Owner	Opens Dashboard → KRI detail. Reviews current value, trend, and linked risk register entry.
6	**User Action**	Owner	Clicks "Acknowledge". Records acknowledgement with timestamp.

7	**User Action**	Owner	Clicks "Escalate to Incident". Creates risk incident record. Assigns to Finance Manager.
8	**User Action**	Finance Manager	Opens incident. Reviews linked data. Creates action plan: accelerate AR collections, defer non-critical spend.
9	**User Action**	Finance Manager	Assigns action tasks in Work module. Sets 5-day review checkpoint.
10	**System**	Risk Module	Monitors KRI recovery. When cash runway returns above threshold: auto-flags incident as "Recovering".

System Events Emitted

Event Topic	Subscriber(s) / Effect
`risk.kri.breach`	Dashboard (alert card), Notification service (push alert), SDO (risk layer update), Audit log
`risk.incident.created`	Work module (action tasks), Risk register (linked incident), Audit log

Success Criteria

- ✓ Breach detected and escalated without manual monitoring.
- ✓ Responsible executives alerted within one event propagation cycle.
- ✓ Incident documented with action plan and owner.
- ✓ KRI recovery tracked automatically.

WF-21 Annual GRC Assessment

Trigger	Compliance Officer runs the organisation's annual ISO 27001 assessment.
Actors	Compliance Officer · System
Modules	Risk / GRC · Work · Legal · Dashboard

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Compliance Officer	Opens Risk & GRC → Assessments → New Assessment. Selects ISO 27001 framework.
2	**System**	GRC Module	Generates assessment questionnaire: all ISO 27001 control clauses, each with description and evidence guidance.
3	**User Action**	Compliance Officer	Works through controls: marks each Compliant / Partially Compliant / Non-Compliant / N/A.
4	**User Action**	Compliance Officer	For each gap: enters remediation plan, assigns owner, sets target date. Attaches evidence from document vault.
5	**System**	GRC Module	Calculates interim compliance score after each control entry. Overall score visible throughout.
6	**User Action**	Compliance Officer	Completes all controls. Reviews gap summary.
7	**System**	GRC Module	Generates assessment report (PDF) with score, gap summary, and remediation roadmap.
8	**Automation**	Work Module	Creates remediation tasks for all non-compliant controls. Assigns to owners with target dates.
9	**Automation**	Dashboard	GRC compliance score widget updated. Remediation progress tracked over time.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`grc.assessment.completed`	Work module (remediation tasks), Dashboard (GRC score update), Document vault (report archive), Audit log

Success Criteria

- ✓ Assessment completed with every control documented.
- ✓ Compliance score calculated and visible on dashboard.
- ✓ All remediation gaps converted to tracked work tasks with owners.
- ✓ Audit-ready report generated and stored in document vault.

WF-22 Security Incident Response

Trigger	A suspected data breach is reported. CISO initiates incident response.
Actors	CISO / IT Manager · HR · Legal · Executive
Modules	Risk / GRC · Security · HR · Legal · Work

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	IT Manager	Receives external report of suspected breach. Opens Risk & GRC → Incidents → New Incident. Type: Security.
2	**User Action**	IT Manager	Enters incident description, scope, affected systems, initial assessment.
3	**System**	Risk Module	Incident classified as High severity (based on data type). Auto-escalates to CISO and CEO.
4	**User Action**	CISO	Activates incident response team. Assigns tasks in Work module: forensic investigation, system isolation, customer notification assessment.
5	**Automation**	Legal Module	Receives high-severity security incident event. Triggers regulatory notification countdown (GDPR: 72-hour clock).
6	**User Action**	CISO	Investigation complete. Determines scope of exposure. Decides notification is required.
7	**User Action**	Legal Officer	Generates breach notification letter from template. Files with regulatory authority. Records in compliance calendar.
8	**User Action**	CISO	Completes incident with root cause and remediation steps. Closes incident. Post-incident review scheduled.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`risk.incident.created`	Notification service (CISO/CEO alert), Legal (GDPR clock start), Work (response tasks), Audit log
`legal.regulatory.notification_filed`	Compliance calendar (notification record), Document vault (filed letter), Audit log

Success Criteria

- ✓ Incident contained, investigated, and documented.
- ✓ Regulatory notification filed within 72-hour GDPR window.
- ✓ All response actions tracked with full audit trail.
- ✓ Post-incident review scheduled. Remediation in work tracker.

Group 6 — Finance Close

WF-23 Month-End Close

Trigger	Finance Manager performs the monthly period close process.
Actors	Finance Manager · Accountant
Modules	Finance · Legal · Risk · Dashboard

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Finance Manager	Opens Finance → Period Management → [Current Month]. Runs pre-close checklist.
2	**System**	Finance Module	Pre-close checklist: lists unreconciled bank accounts, unposted entries, unpaid invoices outstanding, open accruals.
3	**User Action**	Accountant	Posts any outstanding accruals and prepayments. Reconciles all bank accounts. Resolves all checklist items.
4	**User Action**	Finance Manager	Reviews period-end P&L and Balance Sheet. Queries any unexpected variances.
5	**User Action**	Finance Manager	Satisfied with period data. Clicks "Close Period". Confirms warning: no further posting after close.
6	**System**	Finance Module	Period locked. All posting attempts to this period will now be blocked.
7	**Automation**	Finance Module	Generates period-end reports: P&L, Balance Sheet, Cash Flow Statement. Stores as immutable snapshots.
8	**Automation**	Finance Module	Emits finance.period.closed. DAO smart contracts receive event to evaluate royalty payment triggers.
9	**Automation**	Analytics	Updates MoM variance analysis. Flags significant movements for management attention.
10	**Automation**	Dashboard	Finance dashboard refreshes with closed-period actuals. MoM charts updated.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.period.closed`	DAO smart contracts (royalty evaluation), Analytics (MoM update), Dashboard (chart refresh), Risk (update financial KRIs)
`finance.reports.generated`	Document vault (immutable snapshot), Dashboard, Audit log

Success Criteria

- ✓ Period closed. All transactions locked.
- ✓ Period-end reports generated and stored as immutable snapshots.
- ✓ DAO royalty smart contracts triggered where applicable.
- ✓ MoM variance analysis available for management review.

WF-24 Consolidated Group Financial Reporting

Trigger	Group CFO requires consolidated P&L and Balance Sheet across all group entities at month end.
Actors	CFO · Finance Directors (per entity)
Modules	Finance · DAO · Dashboard · Analytics

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**System**	Finance (each entity)	Each entity OrgOS completes period close. finance.period.closed events emitted to group DAO event bus.
2	**User Action**	CFO	Opens Finance → Consolidated Reports. Selects reporting group (all entities). Selects period.

3	System	Finance Module	Aggregates financial data from all entity OrgOS instances via DAO data sharing layer.
4	System	Finance Module	Applies intercompany elimination rules: removes intercompany sales, loans, and dividends from consolidated view.
5	System	Finance Module	Applies currency conversion if multi-currency group (using configured exchange rates).
6	System	Finance Module	Renders consolidated P&L and Balance Sheet with entity columns and group totals.
7	User Action	CFO	Reviews consolidated statements. Drills into any entity variance. Adds commentary.
8	User Action	CFO	Exports consolidated pack (Excel or PDF) for board distribution.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.consolidated.report_generated`	Document vault (archive), Dashboard (group financial widgets), Audit log

Success Criteria

- ✓ Consolidated P&L and Balance Sheet available within minutes of last entity period close.
- ✓ Intercompany eliminations applied automatically.
- ✓ Board-ready export produced in one click.
- ✓ All entity data timestamped for audit purposes.

WF-25 Year-End and Audit Preparation

Trigger	Owner and Finance Manager prepare for the annual statutory audit.
Actors	Owner · Finance Manager · External Auditor (external)
Modules	Finance · Legal · Risk · Settings / Audit

Workflow Steps

Step | Phase | Actor | Action / System Response

1	User Action	Finance Manager	Completes year-end period close (as per WF-23). All 12 monthly periods locked.
2	User Action	Finance Manager	Opens Finance → Reports → Year-End Pack. Selects financial year.
3	System	Finance Module	Generates year-end pack: full-year P&L, Balance Sheet, Cash Flow, Notes, Trial Balance, audit-ready JE listing.
4	User Action	Finance Manager	Opens Legal → Compliance → Prepare Audit Pack. Scopes to this entity and financial year.
5	System	Legal Module	Compiles: all JEs with attachments, payroll summaries, filed returns, vendor contracts, employee contracts, board resolutions.
6	User Action	Finance Manager	Reviews pack. Annotates any items requiring explanation. Generates final pack.
7	User Action	Owner	Creates a time-limited Auditor OrgExec (AUDITOR-ODD): read-only, scoped to audit period, auto-expires in 30 days.
8	External	External Auditor	Logs in as Auditor OrgExec. Browses audit pack via portal. Raises queries in audit query log.
9	User Action	Finance Manager	Responds to queries in audit query log. Provides additional evidence via document vault.
10	External	External Auditor	Signs off accounts. Files with relevant authority.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.yearend.pack_generated`	Document vault, Audit log
`auth.auditor_orgexec.created`	ODDBus (register auditor ODD), Audit log

Success Criteria

- ✓ Complete year-end data pack available with full audit evidence trail.
- ✓ Auditor given scoped, time-limited read-only access — no over-provisioning.
- ✓ All auditor queries and responses tracked in audit query log.
- ✓ Accounts signed off. Statutory filing completed.

Group 7 — AI Agent Operations

WF-26 AI Scheduling Agent — Daily Operation

Trigger	Scheduling AI OrgExec runs its daily scheduling optimisation for the service business.
Actors	Scheduling AI OrgExec (automated) · Operations Manager (oversight)
Modules	ODD / ODDBus · Operations · CRM · HR

Workflow Steps

Step | Phase | Actor | Action / System Response

1	Automation	Scheduling AI	At 6:00 AM: AI OrgExec wakes per lifecycle policy in ODD. Reads order queue from CRM (permitted by ODD capability).
2	Automation	Scheduling AI	Reads staff availability from HR (permitted by ODD: read HR.schedule, current department only).
3	Automation	Scheduling AI	Reads machine availability from Operations OrgCPU status (permitted by ODD: read OrgCPU.status).
4	Automation	Scheduling AI	Runs optimisation model: assigns orders to machines and staff to minimise turnaround time.
5	Decision	ODDBus	Does the generated schedule change affect >3 staff members simultaneously? (ODD constraint threshold)
6	Automation	ODDBus	IF BELOW THRESHOLD: schedule applied automatically. Staff notified via CommDriver.
7	Automation	ODDBus	IF ABOVE THRESHOLD: ODD Escalate decision. Action paused. Alert sent to Operations Manager.
8	User Action	Operations Manager	Reviews pending schedule in OrgExec console. Approves or modifies.
9	Automation	Scheduling AI	Once approved (or auto-applied): writes schedule assignments to Operations module (ODD write capability used).
10	Automation	Scheduling AI	Notifies each staff member of their assignments via CommDriver (ODD notification capability used).
11	System	Audit Log	All AI actions logged: what was read, what was written, what was notified, and whether escalation occurred.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`odd.action.allowed`	Audit log (all AI read/write actions)
`odd.action.escalated`	Notification (manager alert), ODDBus (action paused), Audit log
`operations.schedule.updated`	Operations dashboard, CommDriver (staff notifications), SDO (OrgExec node update)

Success Criteria

- ✓ Daily schedule optimised and applied with zero manual effort on routine days.
- ✓ Any day where >3 staff are affected: human manager reviews before applying.
- ✓ Full AI action audit trail — every read, write, and notification logged.
- ✓ Staff receive their daily assignments automatically.

WF-27 AI Agent ODD Escalation Handling

Trigger	AI OrgExec encounters a situation outside its ODD constraints and escalates.
Actors	AI Customer Comms Agent · Customer Service Manager (human)
Modules	ODD / ODDBus · CRM · Communications · Work

Workflow Steps

Step | Phase | Actor | Action / System Response

1	Automation	Customer Comms AI	Reads incoming customer complaint message (permitted by ODD: read CRM.messages).
2	Automation	Customer Comms AI	Classifies complaint as: pricing dispute. Confidence: 82%.
3	Decision	ODDBus	Is this action type in the AI agent's ODD capabilities? (Pricing discussions: NOT in capabilities list.)
4	System	ODDBus	ODD Decision: Deny(reason: "pricing_discussion not in capabilities"). Action blocked.
5	Automation	ODDBus	Escalation rule triggered: "Route pricing disputes to Customer Service Manager OrgExec." Escalation event emitted.
6	Automation	Work Module	Creates escalation task assigned to Customer Service Manager: "Customer complaint — pricing dispute. Requires human response."
7	Automation	Customer Comms AI	Sends interim acknowledgement to customer (permitted template): "We've received your message. A team member will respond within 2 hours."
8	User Action	Customer Service Manager	Reviews escalated task. Opens CRM message thread. Reads customer complaint.
9	User Action	Customer Service Manager	Responds to customer directly. Resolves dispute.
10	User Action	Customer Service Manager	Closes escalation task. Adds resolution note.
11	Automation	ODDBus	Escalation closed and resolution recorded in ODD violation/escalation log.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`odd.action.denied`	Audit log, SDO (OrgExec node — escalation count update)
`odd.action.escalated`	Work module (task creation), Notification (manager alert), Audit log
`work.task.completed`	ODDBus (escalation closed), CRM (customer resolution note), Audit log

Success Criteria

- ✓ Customer received timely acknowledgement automatically.
- ✓ Pricing dispute handled by a human — as required by ODD.
- ✓ Zero AI over-reach: agent stayed within ODD boundaries.
- ✓ Full escalation trail in audit log. Manager resolved within 2 hours.

WF-28 AI Agent Performance Review and ODD Update

Trigger	Operations Manager reviews AI Scheduling Agent performance over the past month and tunes its ODD.
Actors	Operations Manager · Admin
Modules	ODD Manager · Analytics · OrgExec Registry

Workflow Steps

Step | Phase | Actor | Action / System Response

1	User Action	Operations Manager	Opens OrgSystem → OrgExec Registry → Scheduling AI profile.
2	System	ODD Manager	Displays agent performance dashboard: tasks completed, escalation rate, ODD violations, actions allowed vs. denied, cost.
3	User Action	Operations Manager	Reviews: escalation rate was 8% this month. Investigates escalation log. Finds most escalations were for 4-staff schedule changes.
4	Decision	Operations Manager	Is the 3-staff threshold causing unnecessary escalations for minor schedule changes?
5	User Action	Operations Manager	Decides to raise threshold to 5 staff. Opens ODD Editor. Updates constraint: max_staff_change_without_approval = 5.

6	**System**	ODD Manager	Shows diff: old constraint (3) vs. new constraint (5). Prompts for change reason.
7	**User Action**	Operations Manager	Enters reason. Saves new ODD version.
8	**System**	ODDBus	Applies graceful transition: in-flight agent actions complete under old ODD. New ODD active from next execution cycle.
9	**System**	OrgDNA Manager	Increments OrgDNA version: updated ODD template recorded.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`odd.contract.updated`	ODDBus (new version active), OrgDNA Manager (config update), Audit log
`orgdna.version.created`	OrgDNA vault, Audit log

Success Criteria

- ✓ AI agent ODD updated. Escalation rate expected to reduce from 8% to ~3%.
- ✓ ODD change fully documented: who changed, what changed, why, when.
- ✓ OrgDNA updated. Config history maintained.
- ✓ Agent continues operating without interruption.

Group 8 — DAO & Franchise Operations

WF-29 Form a Franchise DAO

Trigger	Bob and Alice want to franchise their dry cleaning brand. They form a DAO for franchisee governance.
Actors	Franchise Owner (Bob & Alice) · First Franchisee (external)
Modules	DAO & DLT · OrgDNA Manager · Legal · Finance

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Bob & Alice	Opens DAO & DLT → Form New DAO. Enters: "Premier Dry Cleaning Network", governance: majority vote, quorum: 60%.
2	**User Action**	Bob & Alice	Uses Smart Contract Builder to define terms: (1) 6% monthly royalty; (2) OrgDNA must stay within 1 major version; (3) governance votes for pricing changes.
3	**System**	DAO Module	Compiles smart contracts from visual builder. Previews legal interpretation of each term.
4	**User Action**	Bob & Alice	Opens OrgDNA Manager. Exports current OrgDNA as "PremierDryCleaning-v1.0.0" — the brand package.
5	**User Action**	Bob & Alice	Back in DAO wizard: publishes OrgDNA as canonical brand package. Confirms DAO formation. Clicks "Launch DAO".
6	**System**	DLT Node	DLT network initialises. Genesis block committed with DAO config, smart contracts, and founding OrgDNA v1.0.0.
7	**User Action**	Bob & Alice	Invites first franchisee by email (or UME Org ID).
8	**External**	Franchisee Admin	Receives invitation. Reviews smart contract terms. Accepts. Multi-signs.
9	**Automation**	DLT Node	ORGOS_JOIN transaction committed. Franchisee OrgOS connected to DAO inter-org event bus.
10	**System**	DAO Dashboard	Member network graph shows 2 nodes (Bob & Alice HQ + Franchisee). Treasury and ledger visible.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`dao.formed`	DLT genesis block commit, DAO dashboard initialise, Audit log
`dao.member.joined`	DAO member network update, DLT (ORGOS_JOIN block), Smart contract engine (activate for this member), Audit log
`orgdna.brand_package.published`	All DAO members (broadcast notification), DLT (OrgDNA version record)

Success Criteria

- ✓ Franchise DAO live with DLT network active.
- ✓ Smart contracts deployed: royalty, brand standards, governance rules.
- ✓ OrgDNA brand package published. Franchisees can clone it.
- ✓ First franchisee joined. Inter-org event bus connected between HQ and franchisee.

WF-30 Monthly Royalty Smart Contract Execution

Trigger	Franchisee completes month-end close. Royalty smart contract fires automatically.
Actors	System (fully automated) · Finance Module (both orgs)
Modules	Finance · DAO / DLT · Smart Contract Engine

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**Automation**	Franchisee Finance	Franchisee Finance Manager closes April period. finance.period.closed event emitted on franchisee OrgOS.

2	Automation	DAO Event Bus	finance.period.closed event routed to smart contract engine via DAO inter-org event bus.
3	System	Smart Contract Engine	Evaluates royalty term: reads franchisee April gross revenue (authorised by DAO data sharing ODD: read Finance.revenue).
4	System	Smart Contract Engine	Calculates: £28,450 gross revenue × 6% = £1,707 royalty due.
5	System	Smart Contract Engine	Initiates treasury transfer: debit franchisee treasury £1,707, credit parent (Bob & Alice HQ) treasury £1,707.
6	Automation	Franchisee Finance	Royalty expense JE auto-posted on franchisee OrgOS: Royalty Expense Dr £1,707 / Royalty Payable Cr £1,707.
7	Automation	Parent Finance	Royalty income JE auto-posted on parent OrgOS: Royalty Receivable Dr £1,707 / Royalty Income Cr £1,707.
8	System	DLT Node	ROYALTY_PAYMENT block committed. Immutable record: franchisee, parent, period, amount, contract ID, block hash.
9	Automation	Notification	Both organisations notified: "April royalty of £1,707 processed automatically."

System Events Emitted

Event Topic	Subscriber(s) / Effect
`finance.period.closed`	DAO Smart Contract Engine (royalty trigger evaluation)
`dao.smart_contract.royalty_executed`	Franchisee Finance (royalty JE), Parent Finance (income JE), DLT (ROYALTY_PAYMENT block), Notification (both orgs), Audit log

Success Criteria

- ✓ Royalty paid automatically — no invoice, no chasing, no manual transfer.
- ✓ Both organisations' Finance modules updated with correct entries.
- ✓ Immutable DLT record of payment. Independently verifiable.
- ✓ No human intervention required from period close to royalty receipt.

WF-31 OrgDNA Update Broadcast to Franchisees

Trigger	Bob & Alice update their operational processes and want all franchisees to adopt the update.
Actors	Bob & Alice (Parent Org) · Franchisee Admins
Modules	OrgDNA Manager · DAO & DLT · Notification

Workflow Steps

Step | Phase | Actor | Action / System Response

1	User Action	Bob & Alice	Makes changes to their OrgOS: new compliance pack for updated regulations, revised ODD template for cleaning staff.
2	Automation	OrgDNA Manager	Config change triggers OrgDNA v1.3.0 on parent OrgOS. Change summary auto-generated.
3	User Action	Bob & Alice	Opens OrgDNA Manager → Broadcast Update. Selects "PremierDryCleaning-v1.3.0". Marks as mandatory with 14-day cure period.
4	User Action	Bob & Alice	Writes release note explaining the changes. Clicks "Broadcast to DAO".
5	System	DAO Module	Sends OrgDNA update notification to all member OrgOS instances via DAO inter-org event bus.
6	Automation	Each Franchisee OrgOS	Each franchisee admin receives notification: "New OrgDNA update v1.3.0 available. Mandatory. Apply by [date]."
7	User Action	Franchisee Admin	Reviews update diff. Sees: new compliance pack added, one ODD template updated. No breaking changes.
8	User Action	Franchisee Admin	Clicks "Accept Update". System applies OrgDNA diff. Validates. Activates.
9	System	DLT Node	ORGDNA_UPDATE acceptance committed to DLT for this member. Parent can see all member acceptance status.
10	Automation	Smart Contract Engine	Brand standards check: all accepting members now on v1.3.0. Compliant. No cure period triggered.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`orgdna.broadcast.sent`	All member OrgOS instances (notification), DLT (broadcast record), Audit log
`orgdna.update.accepted`	DLT (ORGDNA_UPDATE block per member), Brand standards smart contract (compliance check), Audit log

Success Criteria

- ✓ All franchisees updated to latest brand standards OrgDNA within 14-day window.
- ✓ Acceptance tracked per member on DLT.
- ✓ Brand standards compliance maintained across entire network.
- ✓ Update history maintained for all member OrgOS instances.

WF-32 DAO Governance Vote — Network-Wide Price Change

Trigger	A franchisee proposes raising minimum service prices across the network. A governance vote is held.
Actors	Proposing Franchisee · All DAO Members · Smart Contract Engine
Modules	DAO & DLT · Governance Engine · Notification

Workflow Steps

Step | Phase | Actor | Action / System Response

1	**User Action**	Franchisee	Opens DAO & DLT → Governance → New Proposal. Enters: "Raise minimum press price from £2.50 to £2.75 — cost inflation adjustment."
2	**User Action**	Franchisee	Attaches supporting analysis (cost data from their Finance module). Sets proposed effective date.
3	**System**	Governance Engine	Creates proposal on DLT. Opens 7-day voting period (per governance smart contract). Notifies all DAO members.
4	**Automation**	Notification	All member organisations receive: "New governance proposal: Price Update. Vote by [date]."
5	**User Action**	Each Member Org Admin	Reviews proposal and attached analysis on DAO governance portal. Adds comments to discussion thread.
6	**User Action**	Each Member Org Admin	Casts vote: For / Against / Abstain. Vote recorded on DLT.
7	**System**	Governance Engine	After 7 days: tallies votes. Result: 5 For, 2 Against. 71% majority. Quorum met (6 of 7 voted). APPROVED.
8	**Automation**	Governance Engine	GOVERNANCE_VOTE block committed to DLT with final tally. Result published to all members.
9	**Automation**	Smart Contract Engine	Governance approval triggers OrgDNA update: minimum price configuration updated to £2.75.
10	**Automation**	OrgDNA Manager	Price change packaged into OrgDNA update broadcast to all members.

System Events Emitted

Event Topic	Subscriber(s) / Effect
`dao.proposal.submitted`	DLT (proposal record), Notification (all members), Governance Engine (voting period start)
`dao.vote.cast`	DLT (per-member vote block), Governance Engine (running tally)
`dao.vote.completed`	DLT (GOVERNANCE_VOTE block), Smart contract (execute approved action), All members (result notification)
`orgdna.update.broadcast`	All member OrgOS instances (price config update), DLT (OrgDNA update record)

Success Criteria

- ✓ Network-wide price change decided collectively, transparently, and automatically applied.
- ✓ Every vote recorded on DLT — permanently auditable.
- ✓ Approved change automatically packaged and broadcast to all members via OrgDNA.

✓ No head office decree needed — governance operated by the network itself.